

Creality Pika Software Manual

Installation

Software

<https://www.crealitycloud.com/software-firmware/software/creality-scan> [↗](#)

Firmware

<https://www.crealitycloud.com/software-firmware/other/type-26> [↗](#)

(* Normally, you don't need to download the firmware manually. You can click **Install** in the prompt of the latest version of the software. Only if the installation fails, you need to try to install the firmware manually.)

Windows 10/11 (64-bit)

Recommended configuration: CPU i7 Gen13 and above, Nvidia RTX3060 graphics card and higher(8GB VRAM), 32GB RAM or above.

Minimum configuration: CPU Intel i7 Gen7 and above, NVIDIA Dedicated GPU (6GB VRAM), ≥16GB or above.

Software version: Creality Scan ≥ v4.2.3.

macOS

Recommended configuration: CPU Apple M3 or higher series processor, 32G RAM or above
System version macOS 14 and above.

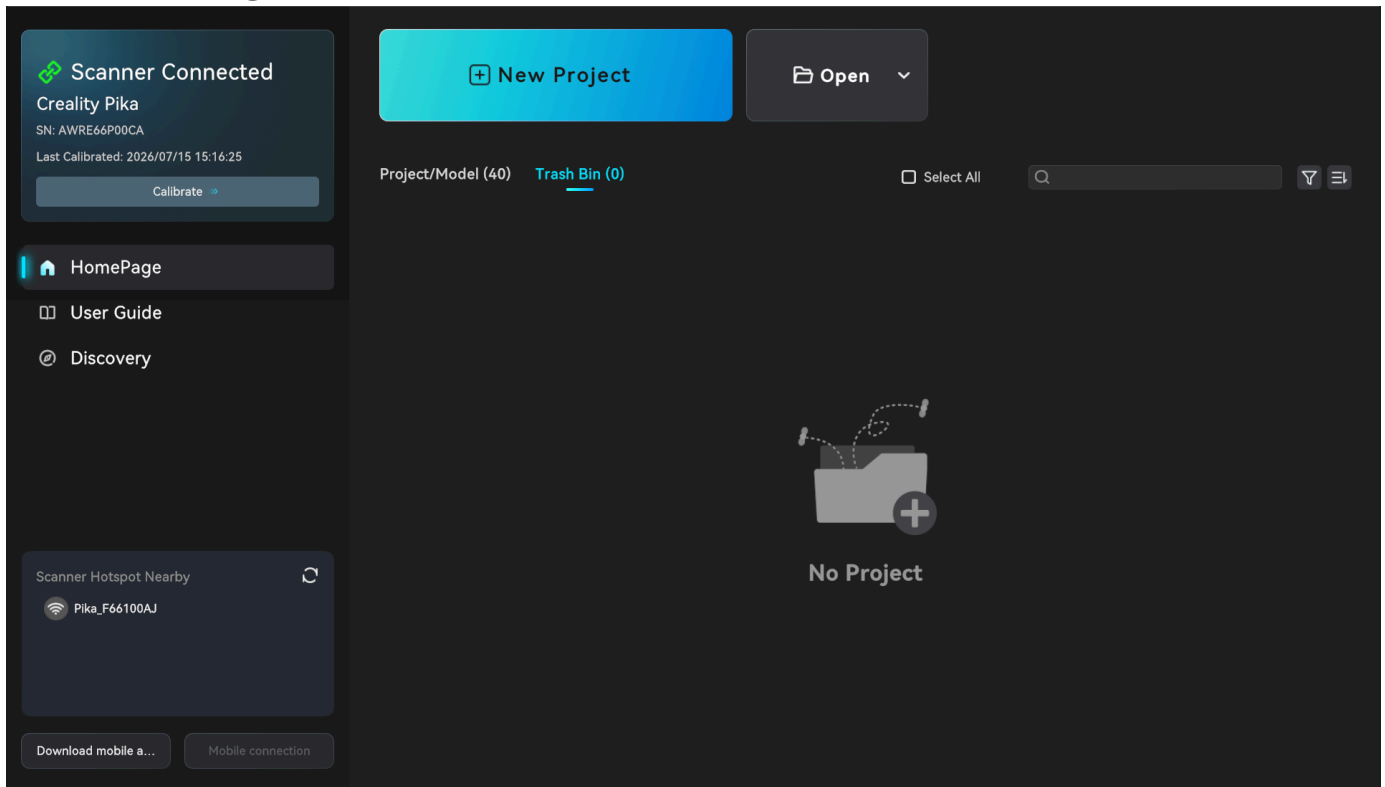
Software version: Creality Scan ≥ v4.2.3.

Pika-series' Recommended Smart Phone Model

Platform	Tier	Recommended / Example Models	Hardware & OS Specifications
Android	Recommended	Samsung :S26 series Xiaomi : 17 series OPPO :Find X9 Ultra Honor Magic 8 Pro	Qualcomm Snapdragon 8 Gen4+, Android 10.0+, 12GB+ RAM
Android	Lowest	Samsung :S23 Ultra Xiaomi :15 series	Android 10.0+, 12GB RAM
IOS	Recommended	iPhone 16 Pro	A17 or later, iOS 15+, 8GB+ RAM

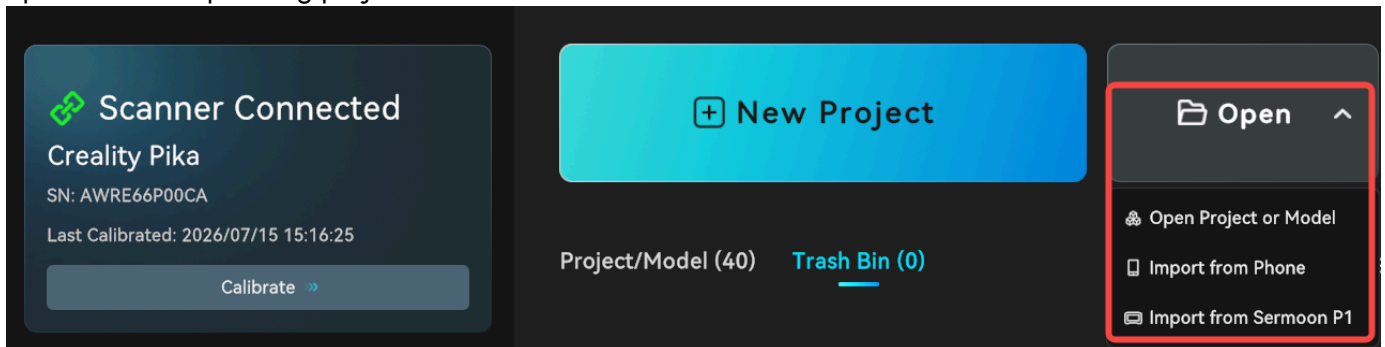
Platform	Tier	Recommended / Example Models	Hardware & OS Specifications
IOS	Lowest	iPhone 14	iOS 15+, 8GB RAM

Home Page



Check whether the scanner is connected successfully in the upper left infobox of the homepage. It is recommended to use USB3.0 to connect the scanner.

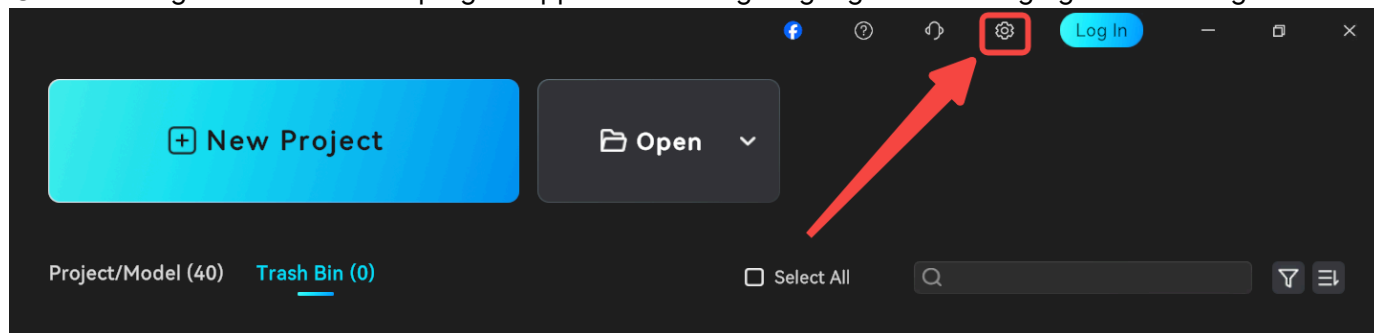
- ① **The left column** shows the basic information of the device at the top, and supports entering the calibration page. "Home", "User Guide", and "Discovery" are displayed below.
- ② **The main area** on the right side lists the recently opened projects. Click the corresponding picture to open the corresponding project.



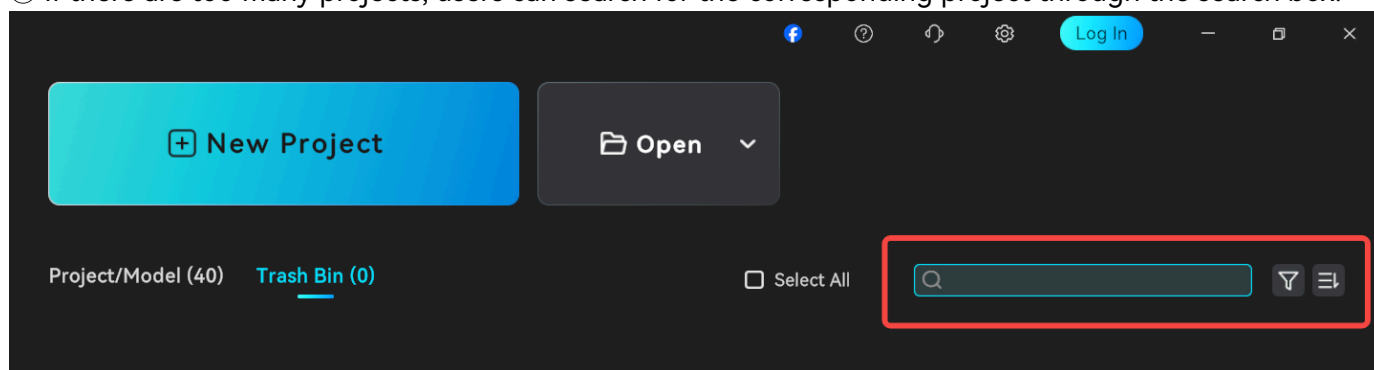
- ③ Click "New Project" at the top to start the scan. Click "Open", there are two options:
 - click "Open Project or Model" to import projects from the local folder (.obp format) or view the model (.stl, .obj, .ply format);

- ▶ click “Import from Phone” to import projects from the "Creality Scan" APP through the LAN, and be sure to keep the mobile phone and computer in **the same LAN**.

④ The settings button on the top right supports switching languages and changing other settings.

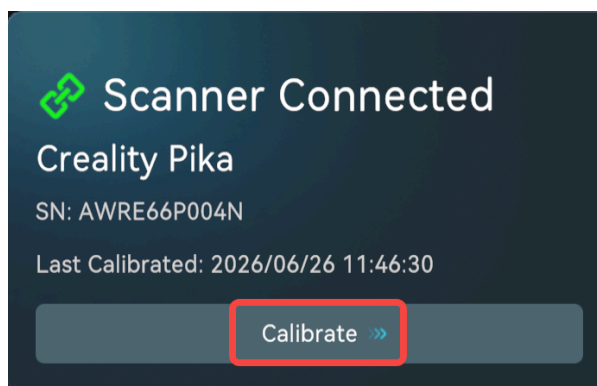


⑤ If there are too many projects, users can search for the corresponding project through the search box.

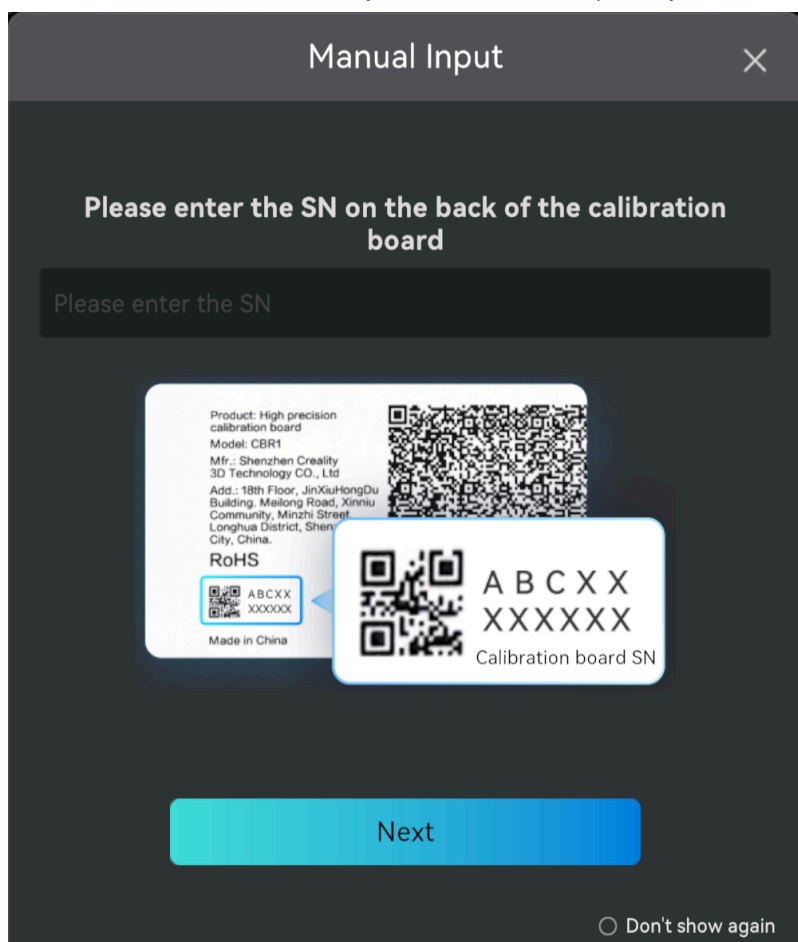


Quick Calibration

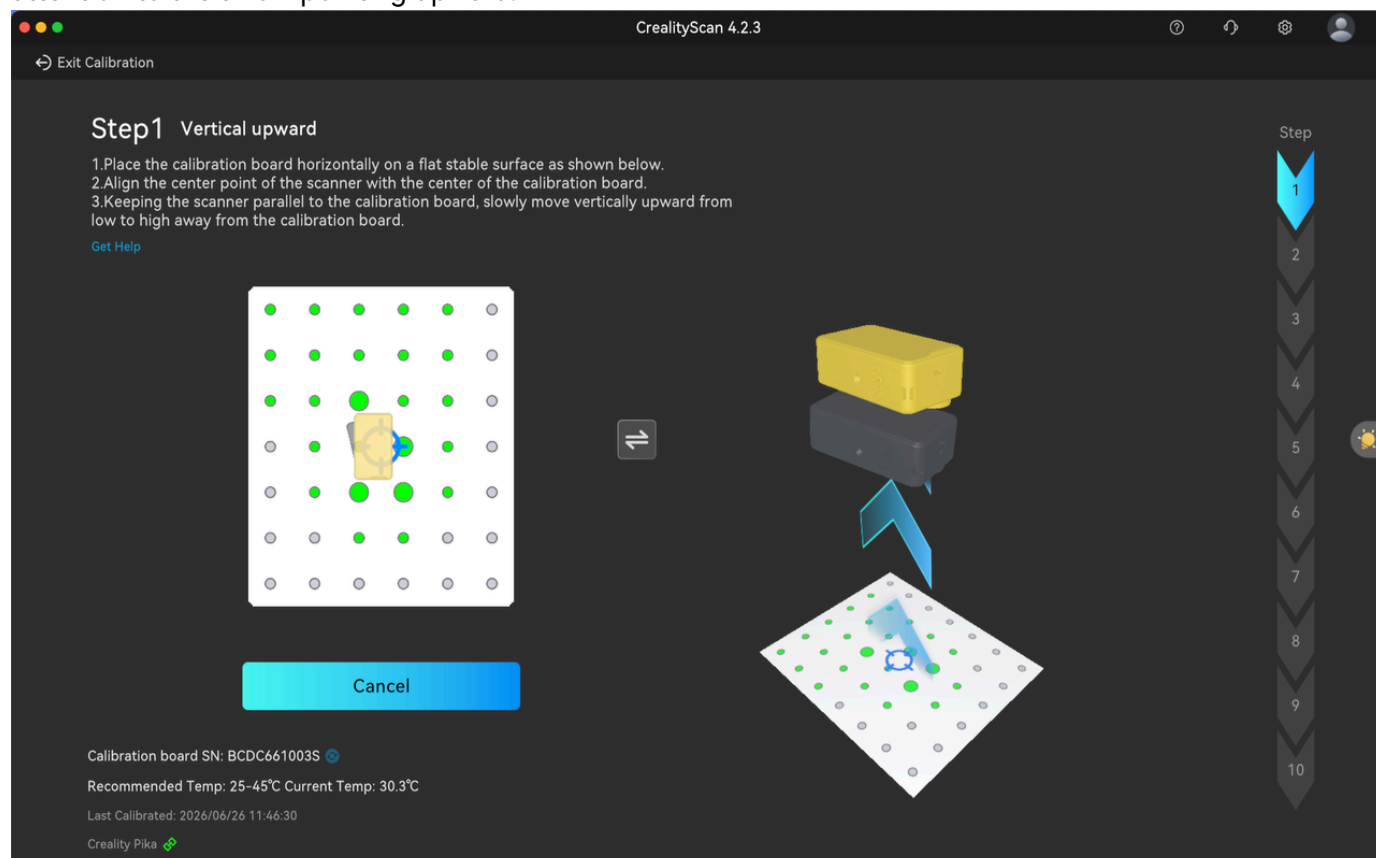
After the device is connected, enter through the calibration button in the upper left infobox of the homepage.



When using it for the first time, please enter the SN code on the back of the calibration board..



After entering the acquisition process, be sure to aim the scanner at the front of the calibration board and move the scanner according to the guidance. When placing the calibration board on the desktop, pay attention to the arrow pointing upward.

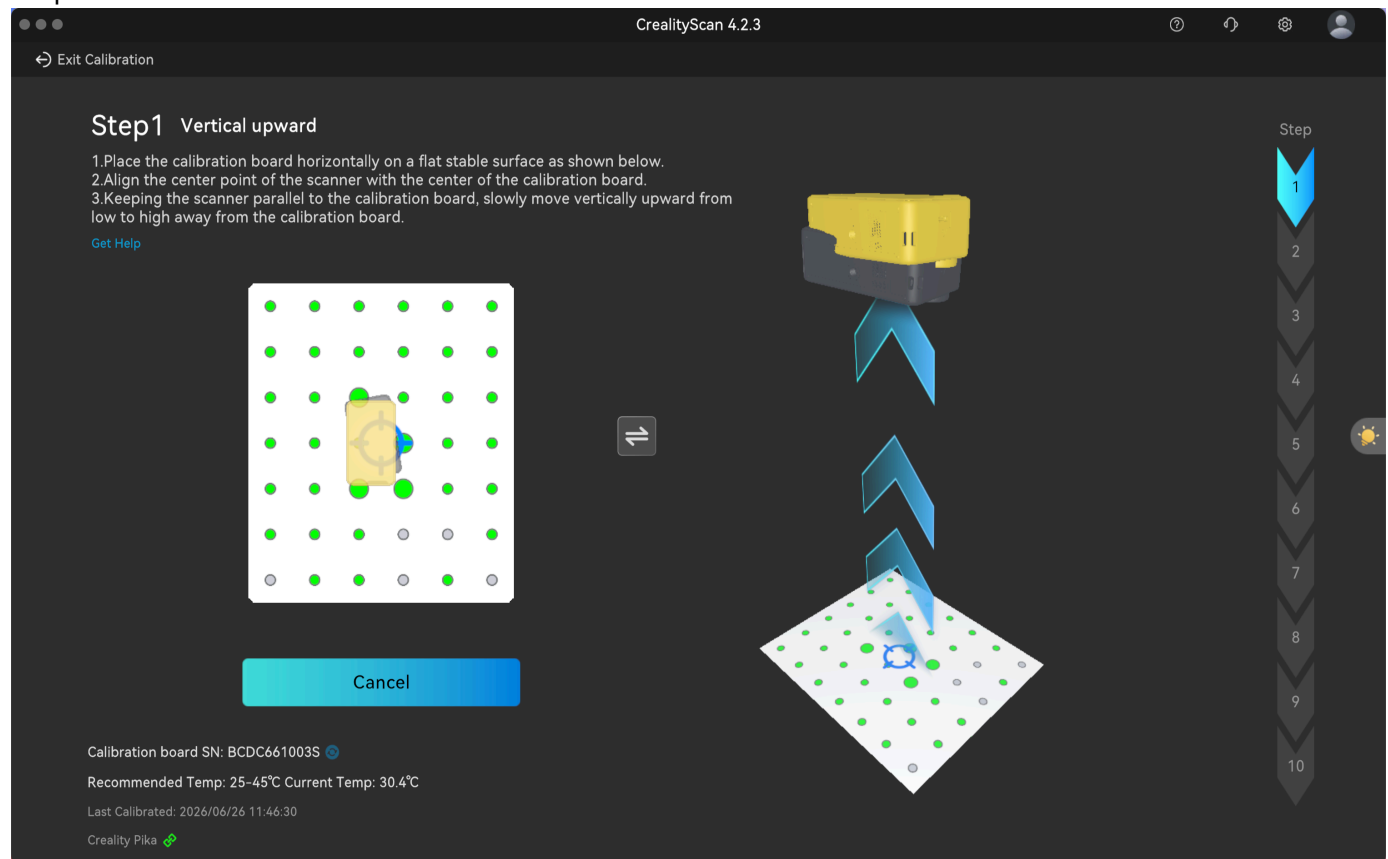


The left side is the process guidance area, the "yellow scanner" displays the pose and angle requirements for the current acquisition. The "black scanner" represents the actual position of the scanner. Pay attention

to matching the direction of the scanner and the calibration board as shown in the illustration.

The bottom of the guidance area displays the calibration board number and scanner temperature. Please make sure to perform calibration within the appropriate temperature range.

The calibration board area on the right side is mainly used to **3D guide** the posture and angle of the scanner in real time. Slowly moves the scanner to the position indicated by the "yellow scanner". The "yellow scanner" will jump to the next height if qualified and maintain the same position unchanged if unqualified.

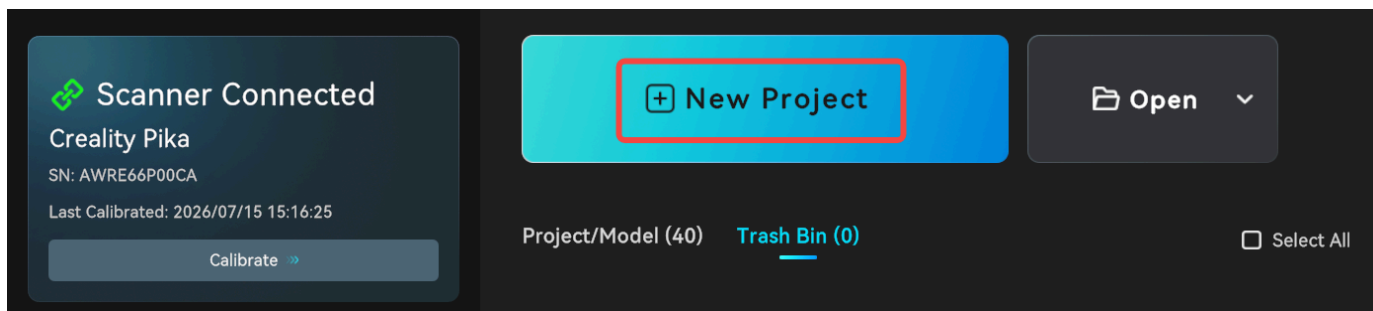


Upon completing all height acquisition of one angle, it will automatically enter the next stage of angle acquisition. Please move the scanner according to the guidance and repeat the entire process to collect all angles.

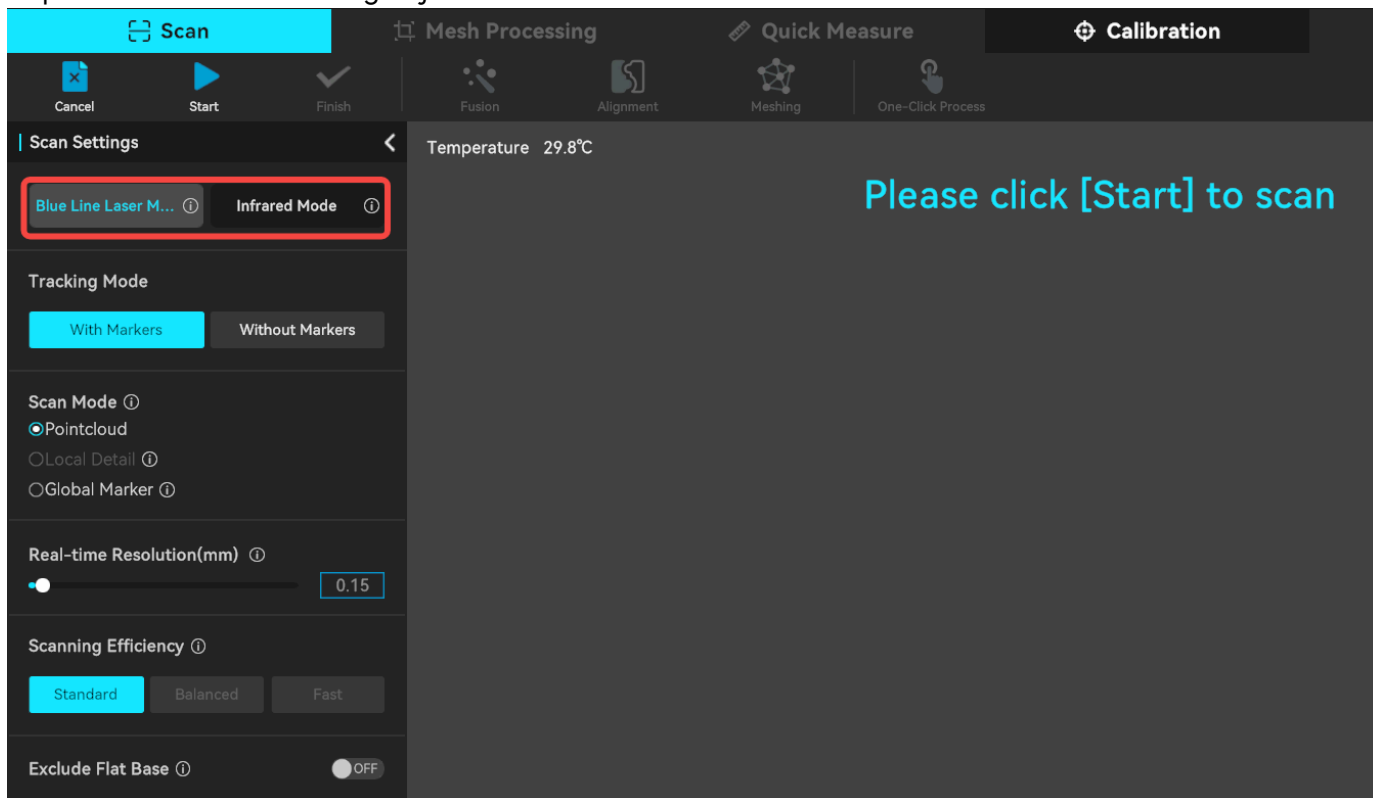
There is a **step indicator bar** on the far right. Upon completing all the steps, it will automatically enter the calibration process. Please wait patiently for the calibration results. After the calibration is successful, please turn on scanning for a high-quality scanning experience.

Scanning Configuration

Click "**New Project**", and in the pop-up window, users can set the project name and project save path. It is recommended to use the name of the scanned object as the project name to increase recognition.

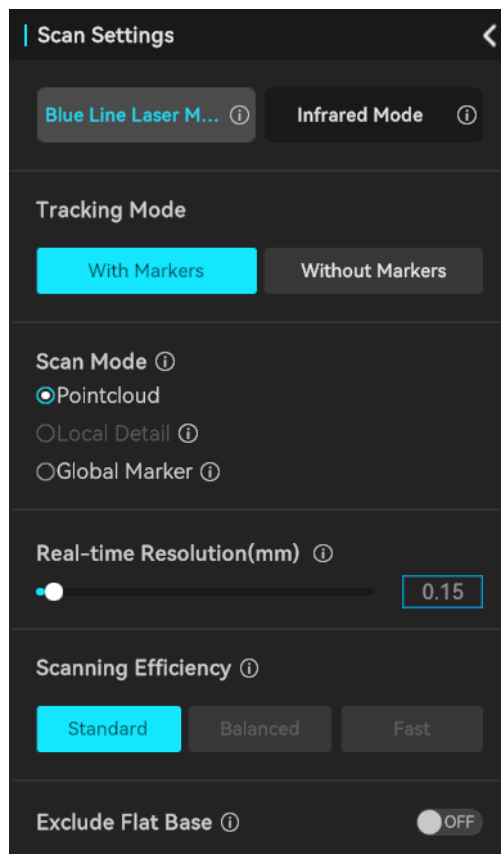


In the "Scan Settings" area of the left sidebar, select the appropriate configuration based on the requirements of the scanning object and environment.



Blue Laser Mode

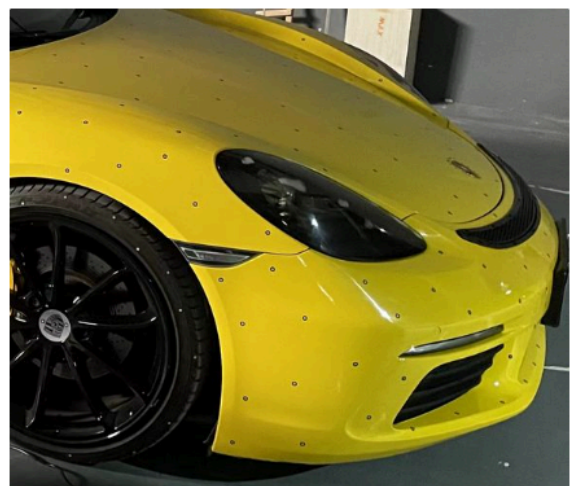
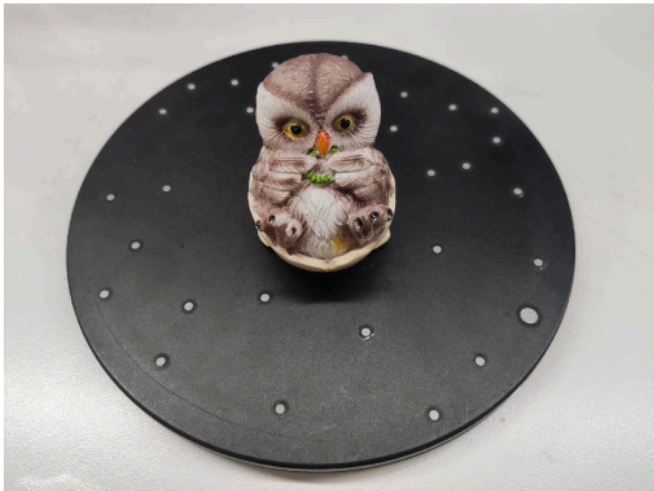
Generally speaking, **blue laser mode** is suitable for scanning objects with rich details and unusual surface materials. It is particularly suitable for industrial manufacturing and inspection scenarios where high precision is required and circular reflective markers are needed for scanning and tracking.



Tracking Mode Selection

Creality Pika's blue line laser scanning mode supports both with markers tracking and without markers(marker-free) tracking.

- ▶ **With Markers tracking** offers higher scanning speed and accuracy compared to markerless tracking, but it requires circular reflective markers for tracking.



- ▶ **Without Markers(marker-free) tracking** does not require markers and works well on medium-to-large objects with rich textures.

Scan Mode Selection

- ▶ **Pointcloud** – Directly captures the full model information of the object.
- ▶ **Local Detail** – Performs a finer pointcloud scan on a selected local area based on the already scanned model.

- **Global Marker** – For medium-to-large objects or objects with few feature points, it is recommended to scan frame points first, followed by the point cloud, to achieve higher model accuracy.

Real-time resolution

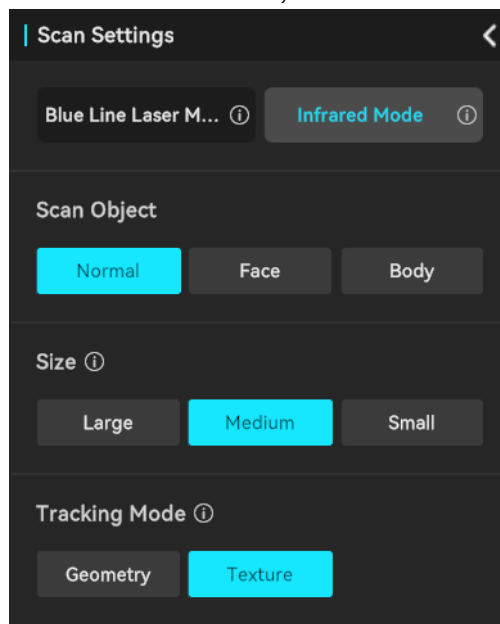
This parameter mainly affects the quality of real-time scan rendering and also influences the final 3D model. A smaller resolution produces a more detailed real-time rendering, but consumes more memory. For smaller objects, a smaller resolution is recommended.

Scanning Efficiency

Adjusts the frame rate during scanning. **Three presets are available: Standard (60 fps), Balanced (90 fps), and Fast (120 fps).** Higher levels enable faster scanning but require higher computer performance. The system will automatically enable available levels based on your hardware.

Infrared Mode

Infrared mode is suitable for fast scanning and supports geometry tracking (without markers, suitable for scanning objects such as human, face, sculpture, and cultural relic), and texture tracking. It excels in face/body scanning (infrared structured light is invisible and harmless to the human eye), medical rehabilitation, digitization of cultural relics and artworks, and scenarios sensitive to ambient light.



Scan Target & Object Size Selection

Select the scan target based on the object being scanned: **Object, Face, or Body.**

Object size can be selected according to the actual dimensions:

- **Large:** 500×500×500 – 2000×2000×2000 mm³
- **Medium:** 250×250×250 – 500×500×500 mm³
- **Small:** 150×150×150 – 250×250×250 mm³

Tracking Mode Selection

(1) **Geometry:** For objects with rich geometric features, it is recommended to use the Geometry Mode for scanning.

(2) **Texture:** For small objects with limited geometric features, it is suggested to apply the Texture Mode after placing marker points. For very small objects, the marker points can be placed on the table. For objects with rich texture but limited geometric features, the Texture Mode can be directly used for scanning.

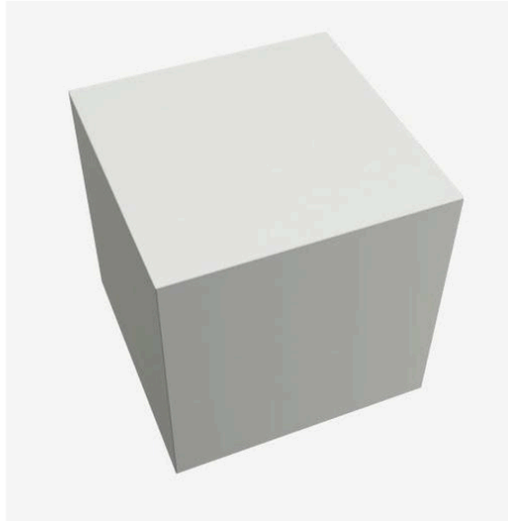
How to determine if an object has rich geometric features?

Objects that have regular geometric shapes, or objects with large flat surfaces, such as cubes, cylinders, cylindrical mugs, square-shaped boxes, or truck cargo compartments often lack sufficient distinctive geometric features and do not have rich geometric features.

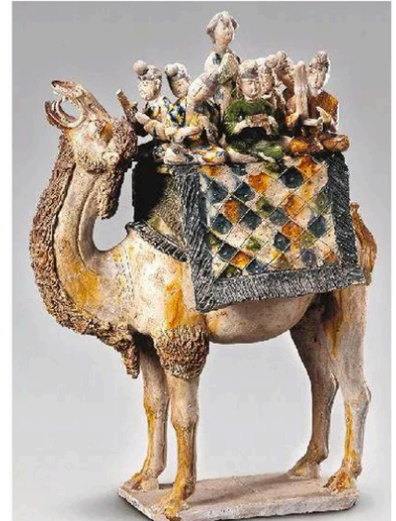
On the other hand, objects like portrait sculptures, shoes, or doll models have rich geometric features.



Rich geometric features



Lack of geometric features
Lack of texture features



Rich texture features

Exclude Flat Base

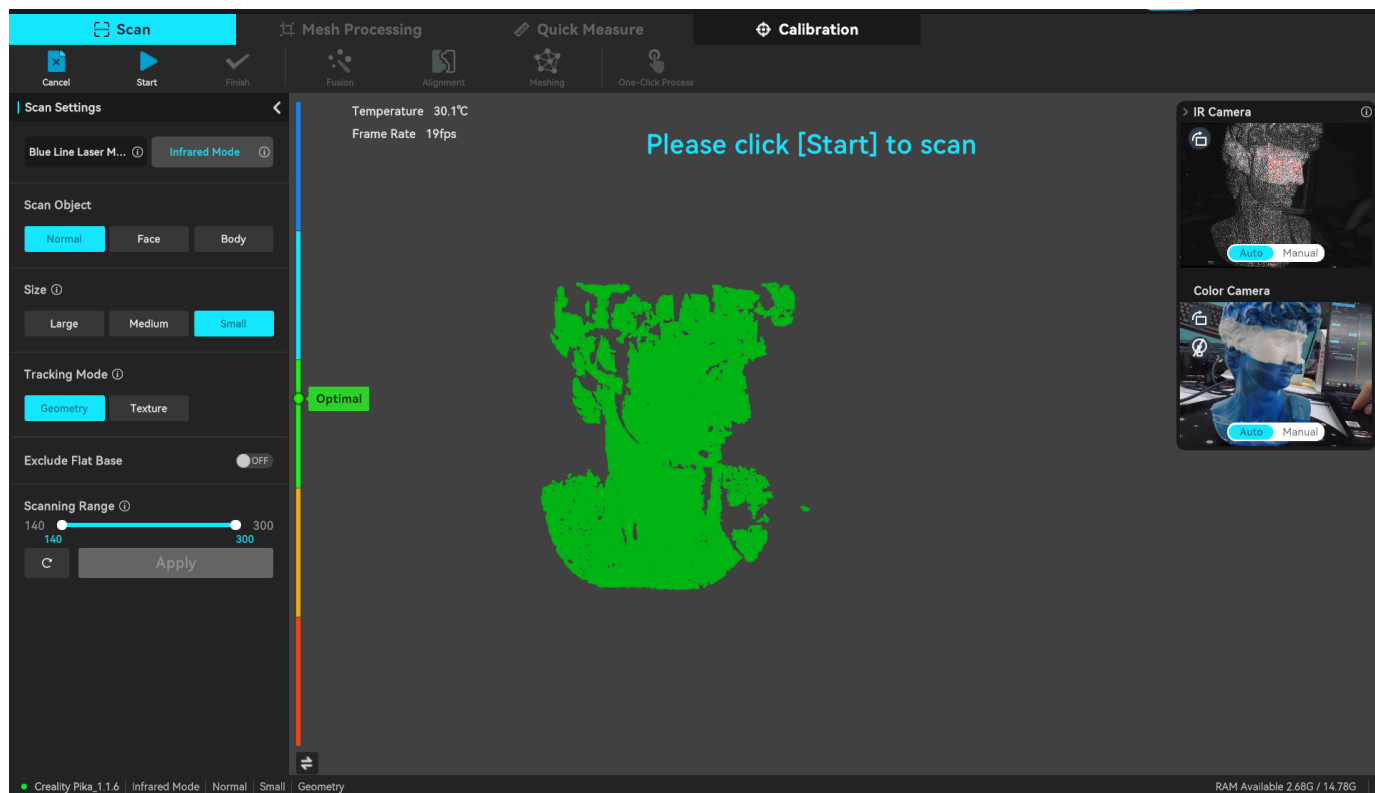
Click **Exclude Flat Base** ☒ to enable or disable it, and the function will take effect in real time. **When enabled**, it can automatically remove planar background detected during scanning

Note:

- It is recommended to enable this function when scanning objects on the table that are small but do not require the assistance of markers.
- When the object to be scanned is extremely small and difficult to scan (e.g., only a few centimeters in size), it is recommended to use the "Marker" mode to scan after pasting markers on the desktop. Do not choose to "exclude flat base" at this time.

Real-time Scanning

After starting a new project, the software will enter the main scanning interface for preview and scanning.

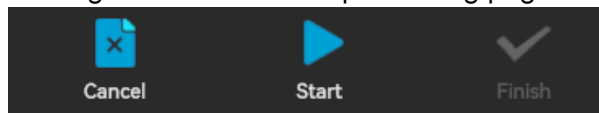


1. Start scanning/Pause scanning/ Resume scanning

Point the scanner at the object you want to scan, and click **"Start"** to begin scanning. Click **"Pause"** to stop scanning.

When resuming after a pause, ensure that the scanner is aimed at the scanned and feature-rich area to quickly return to the state of tracking scanning.

Click **"Finish"** to complete the scanning and will enter the processing page.

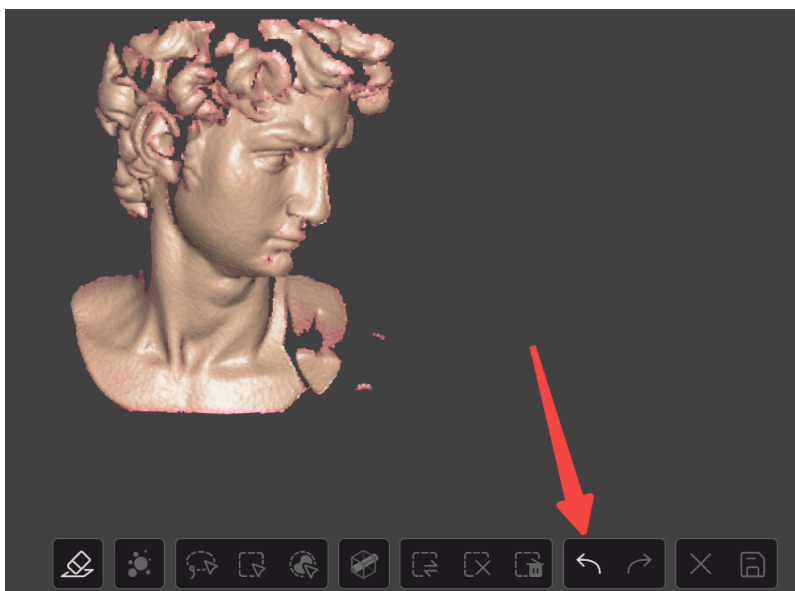


2. Undo/Redo/Cancel/Delete

During the preview stage, if you need to adjust the scan settings, you can click "Cancel" to reset the scan settings and start a new preview.

After starting the scan, click "Delete" to delete the currently scanned data and return to the preview state.

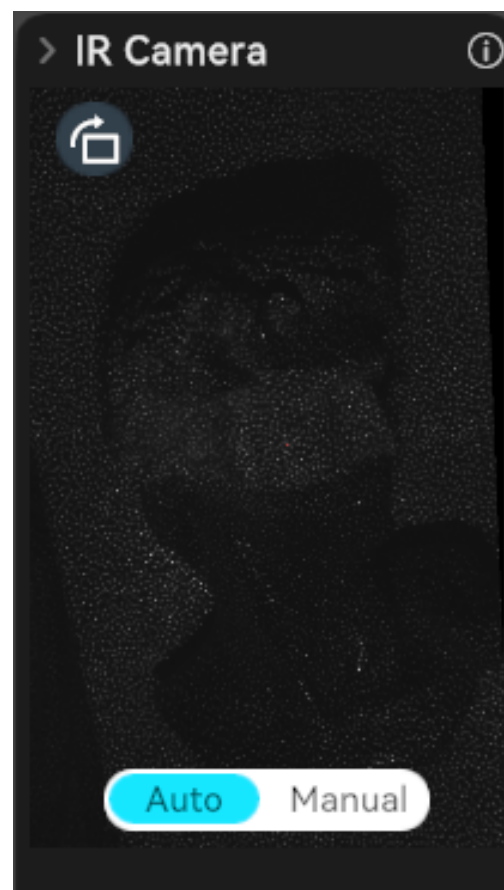
In infrared scanning mode, the software supports scanning undo, allowing you to undo several frames of scanned point cloud data, or redo previously undone frames. This feature is available only when scanning is paused.



3. Depth Camera Preview Window:

Depth Camera Preview Window:

- ▶ The default exposure mode is auto exposure (auto exposure can be used for shooting large objects, human bodies, or faces).
- ▶ If the exposure effect is poor in auto mode, you can click "Manual" to manually adjust the infrared exposure to obtain the best point cloud effect.
- ▶ It is recommended to adjust the image to normal exposure (without blue or red) for better results.
- ▶  Click the rotation icon in the upper left corner of the window to perform a "Rotate right 90°".



RGB Camera Preview Window:

- ▶ The RGB camera window displays the color image of the RGB camera. Click "Manual" to manually adjust RGB exposure and achieve a satisfactory RGB image effect..
- ▶  Click the rotation icon in the upper left corner of the window to perform a "Rotate right 90°".
- ▶  If the environment is dark, click the light bulb icon under the rotate icon to enable the supplemental light for better color mapping.



4. Distance Indicator Bar

Located on the left side of the main scanning interface, when the distance between the scanner and the object being scanned is at the "optimal" distance, the quality of the collected point cloud is optimal, and tracking stability is also optimal.

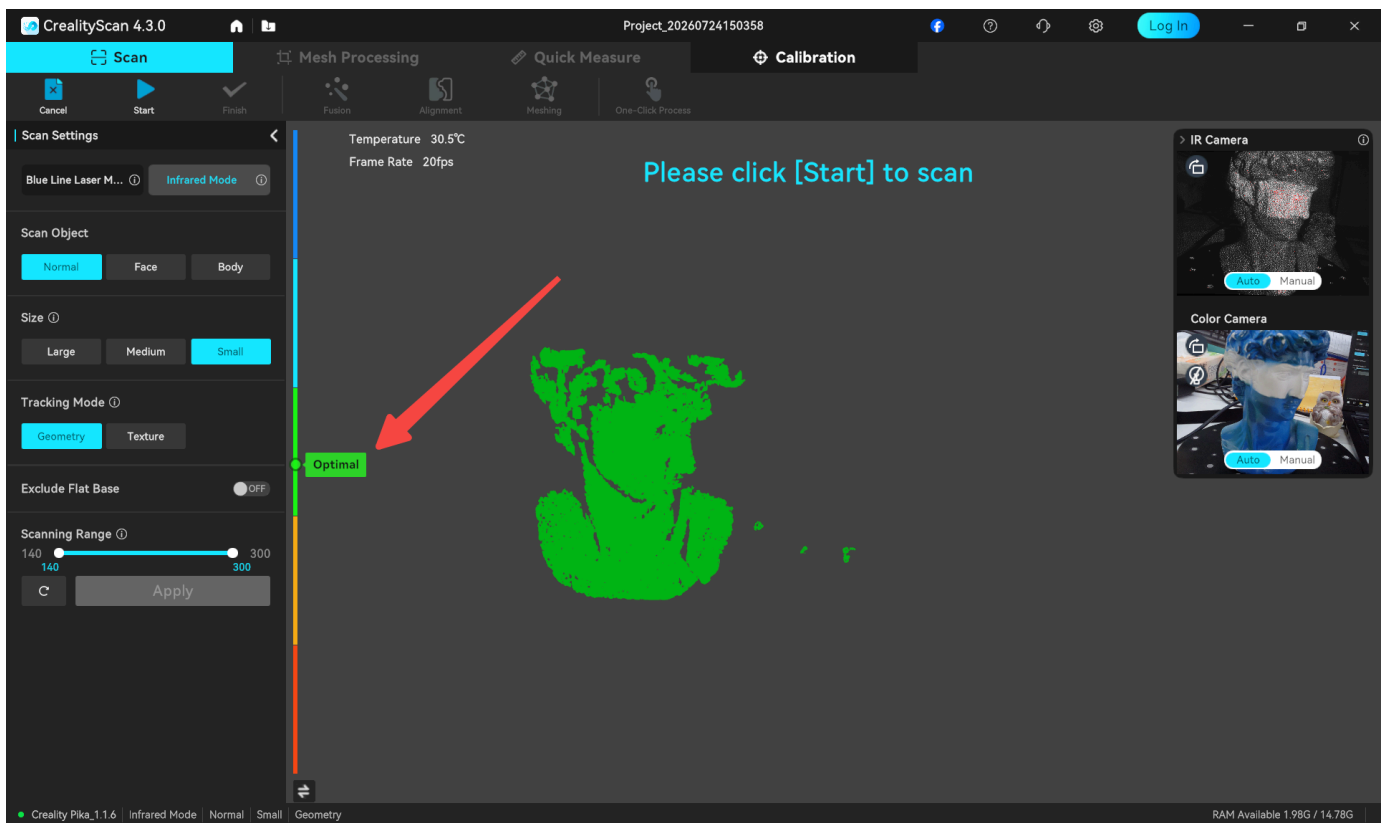


During scanning, please pay attention to the prompts on the distance indicator bar and promptly keep the scanner within the "optimal" distance range.



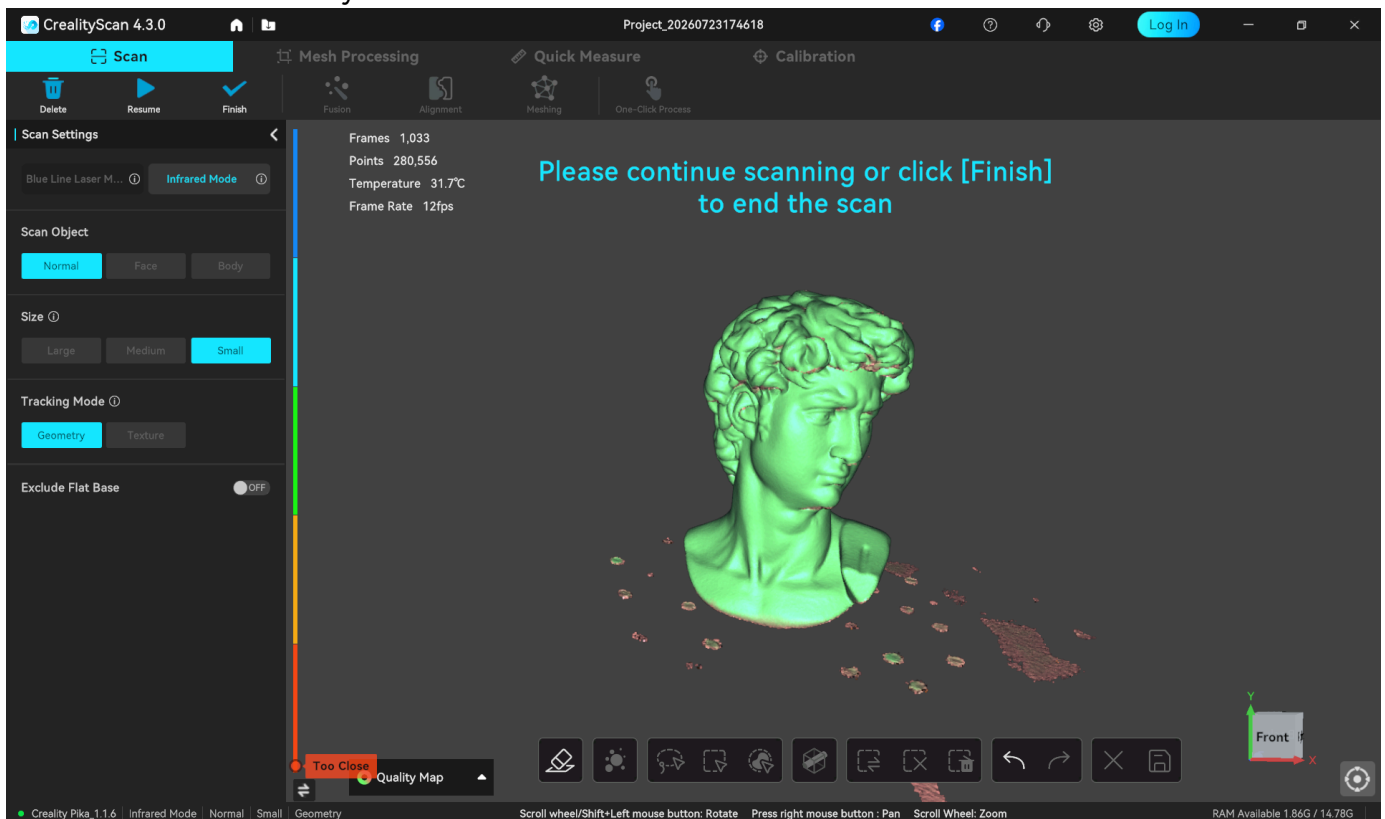
Supports switching the distance indicator bar to cursor mode and histogram mode. It is recommended that beginners use cursor mode, while professional users use histogram mode.

Set the **scanning range** in the left sidebar, adjust the distance range for point cloud collection, and point clouds outside this range will not be collected or displayed in the 3D point cloud rendering area.



5. 3D Point Cloud Rendering Area

The main area is used to preview the current frame point cloud collected by the scanner and the point clouds collected historically.

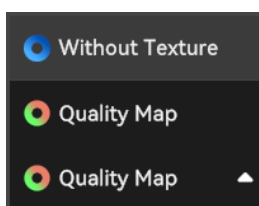


During the pause phase, when the cursor enters the 3D point cloud rendering area, the point cloud can be operated on.

Button	Operation	Remarks
Left mouse button	Hold to select point cloud	Pause scanning
Middle button pressed	Hold and drag to rotate the rendering view	Pause scanning
Middle mouse button scroll	Scroll to zoom in/out the point cloud	Pause scanning
Left mouse button	Hold and drag to pan the point cloud	Pause scanning



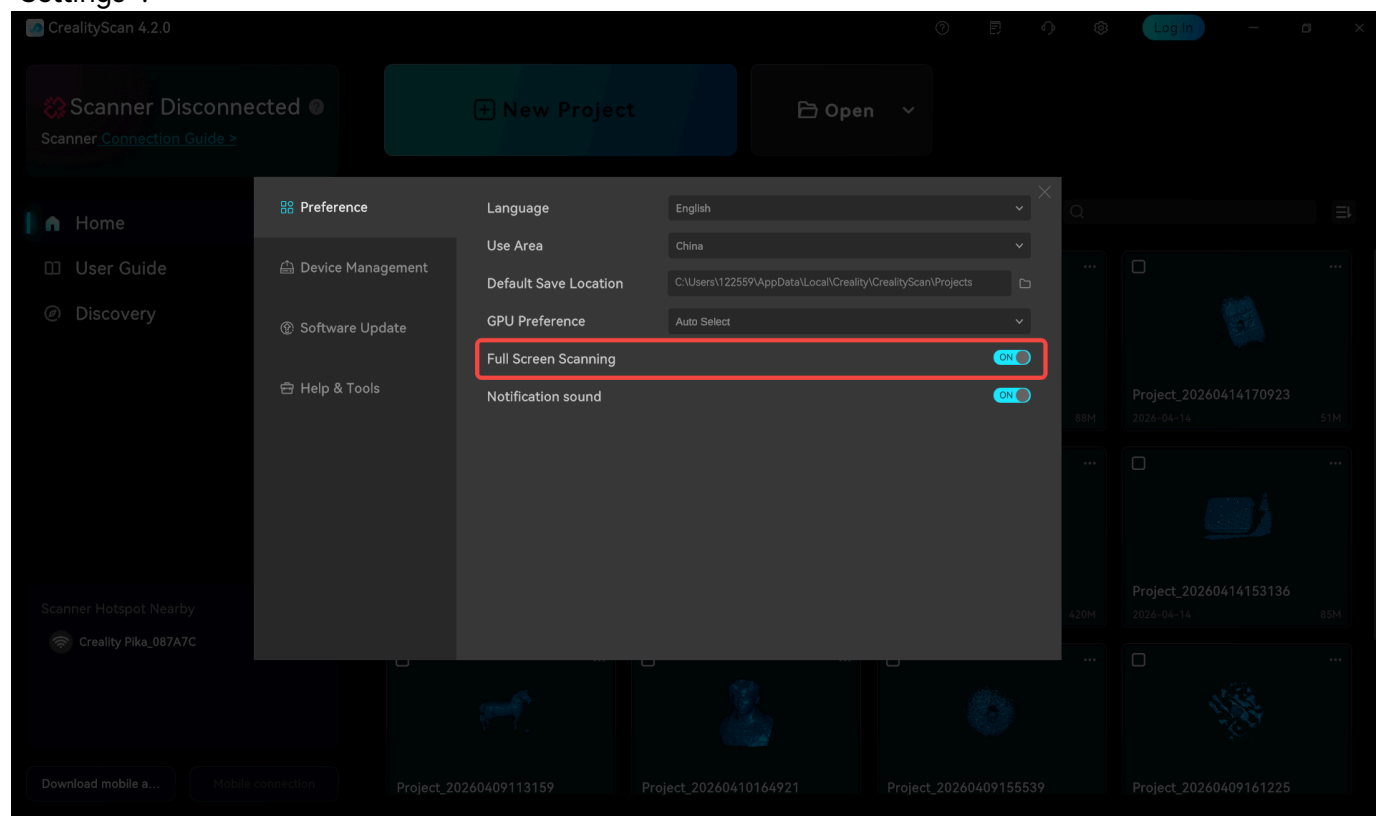
Reset: During the pause phase, after operating on the point cloud, the point cloud can be restored to its state at the time of the pause.



Point cloud rendering mode switching: supporting switching to Without Texture, Quality Map, and With Texture. The default is "Quality Map". In the chromatogram mode, it is recommended to fully scan until the point cloud of the scanned object turns green before clicking "complete scan" to proceed to subsequent processing.

6. Turn off "Full Screen Scanning"

By default, full-screen scanning display is enabled. If you need to see the IR camera preview window and RGB camera preview window during the scanning process, you can turn off "Full Screen Scanning" in "Settings".



7. The current status shows (frames amount, point cloud amount, frame rates, temperature, available memory)

The number of currently collected frames, historically collected point clouds, the current scanner device temperature, and the collection frame rate are displayed in the upper left corner.

On the left side of the bottom status bar, the device connection status and scanning settings are displayed, while on the right side, the available memory is shown.

Blue Laser Mode

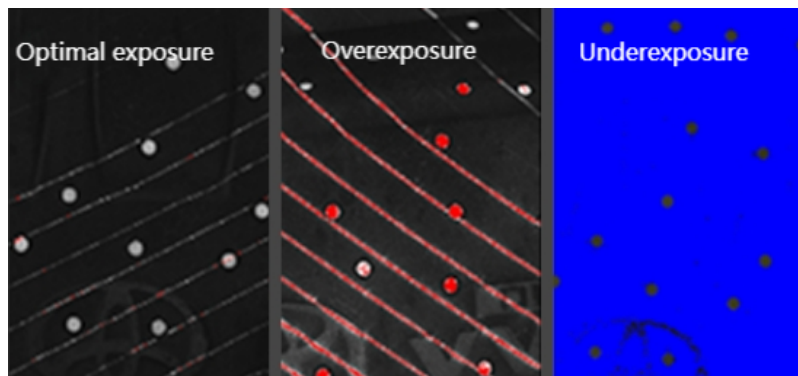
For optimal scanning quality, we recommend using blue laser mode without color mapping.

The main differences from infrared mode are as follows:

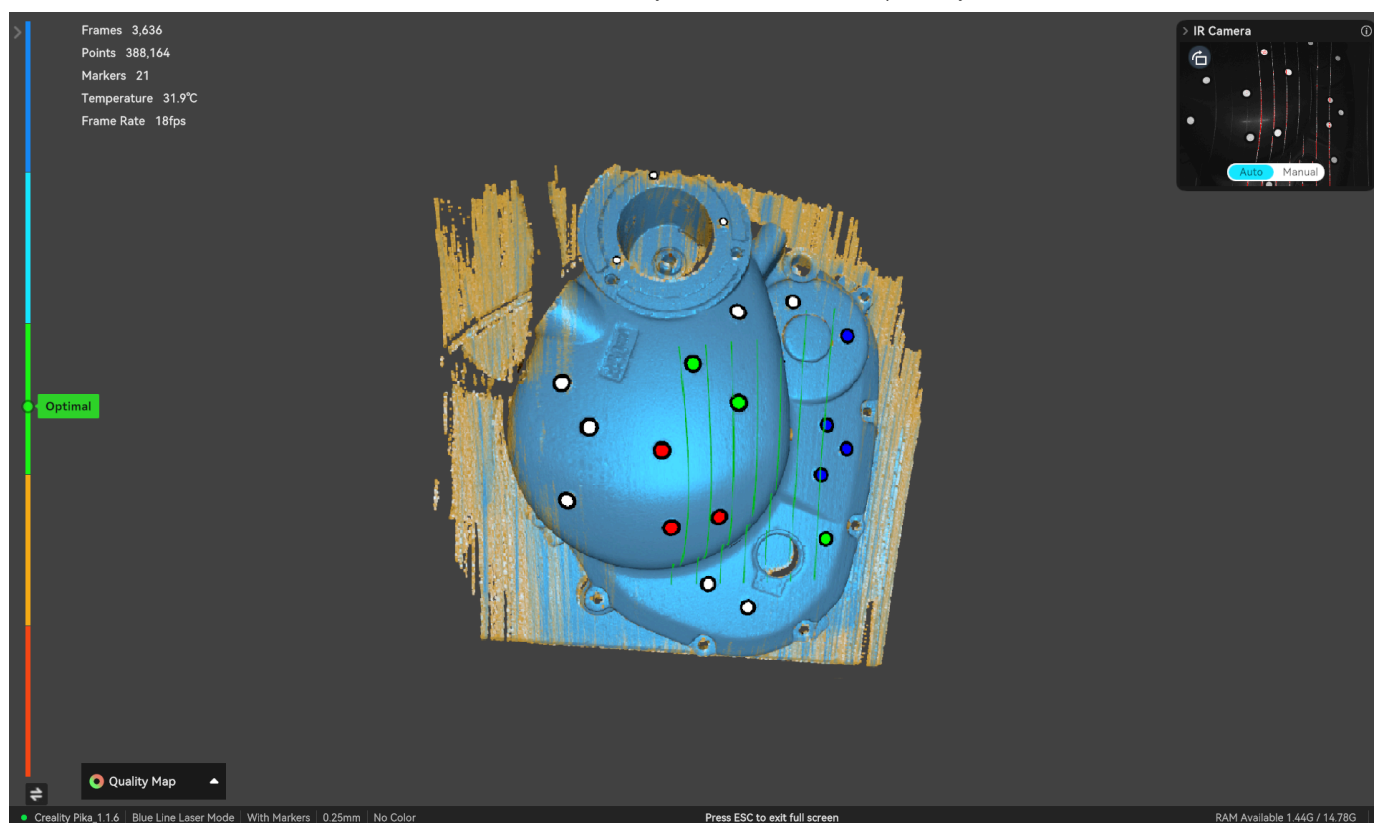
1. In the depth camera preview window, click to expand the menu to adjust the laser brightness. In semi-outdoor scenes, you can manually increase the brightness to reduce sunlight interference. When adjusting, ensure the window is properly exposed (no blue or red).
2. Undo/Redo is not supported.

Blue Laser Scanning Tips:

1. During the preview, adjust the IR exposure and laser brightness to **ensure the depth camera's markers and laser lines are properly exposed** (refer to the picture below. Blue indicates underexposure, red indicates overexposure, and the best state is light white.). Aim the scanner at a feature-rich area of the object and adjust the distance between the scanner and the scanned object to the "optimal distance" according to the distance indicator bar.



2. Click "Start" to start real-time scanning. During the scanning process, while maintaining a relatively fixed distance, slowly move the scanner back and forth towards the scanned part until the point cloud **turns blue (Quality Map)**, then move it back and forth to the unscanned part to continuously ensure normal exposure and "optimal distance".
3. Continue scanning according to the on-screen prompts until all point clouds turn blue.

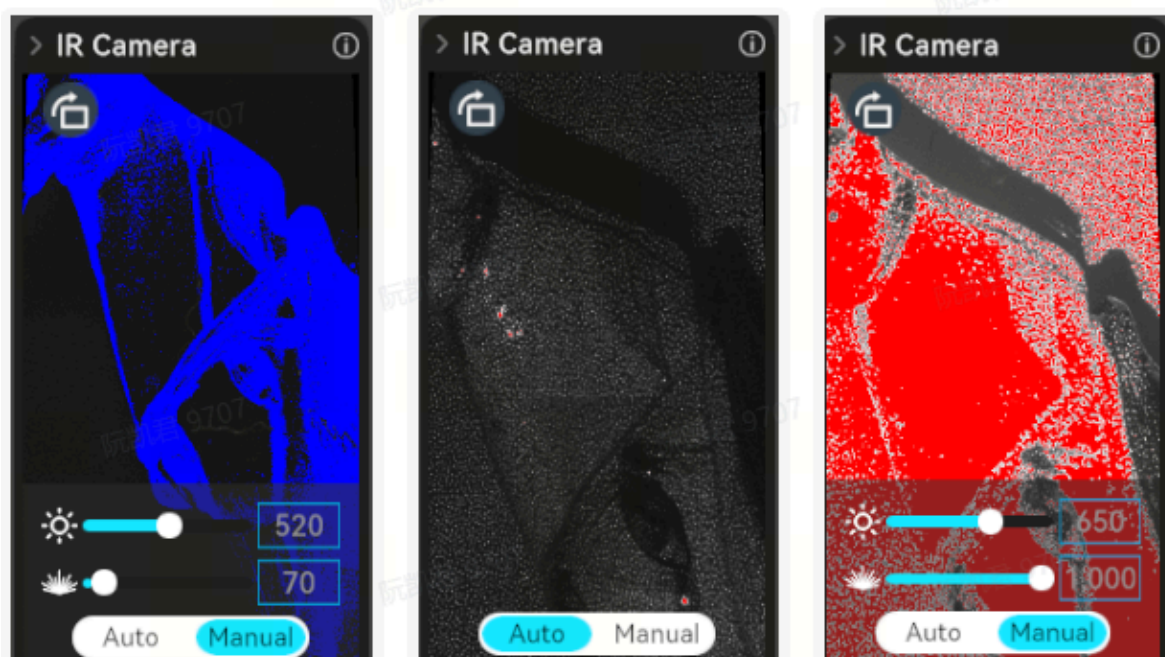


Click "Complete" to end the scan and enter the "Fusion" process.

Infrared Mode

Scanning Tips

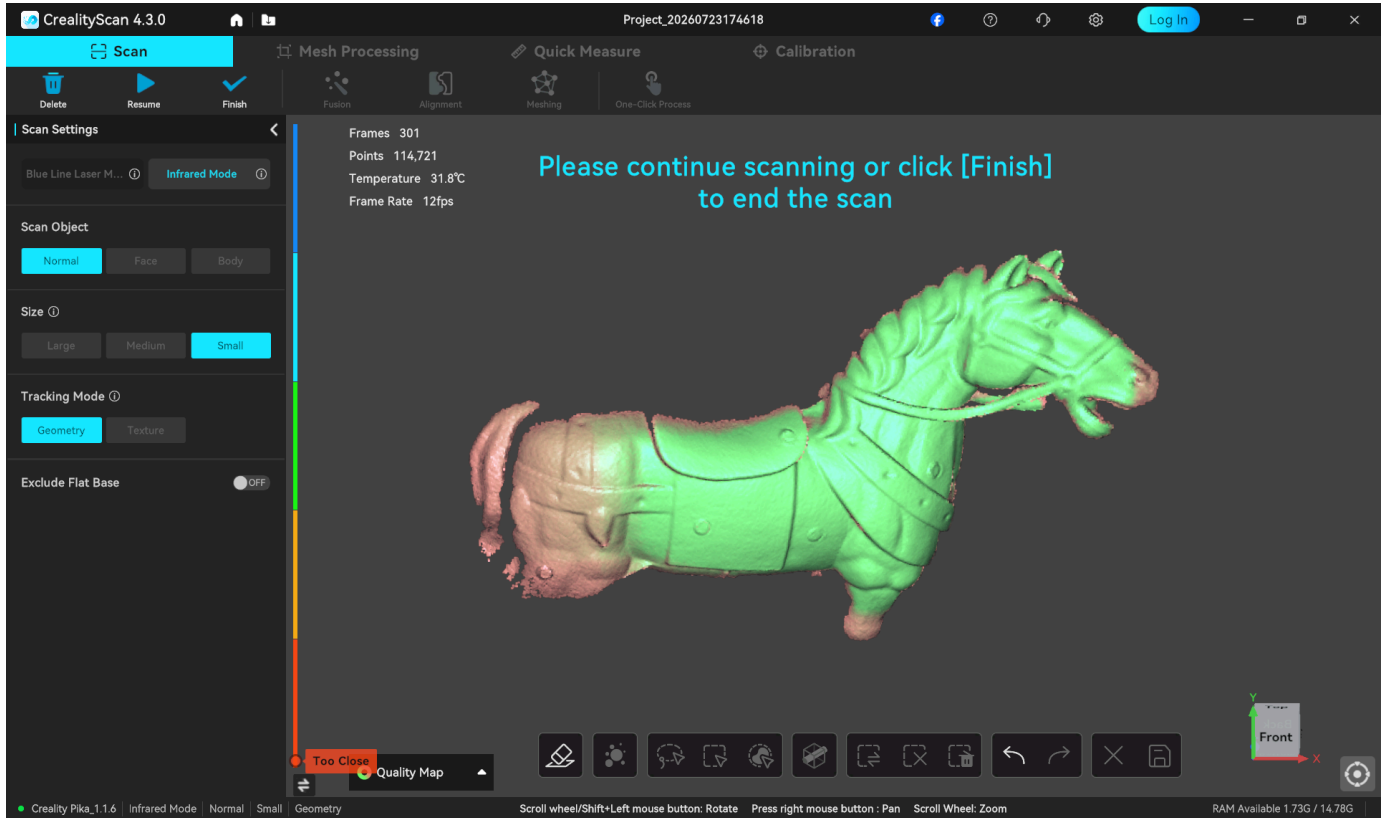
1. When previewing, **ensure that the depth camera preview window is in a normal exposure state** (refer to the picture below. Blue indicates underexposure and red indicates overexposure, and the best state is when there is no blue or red.). Aim the scanner at the feature-rich part of the scanned object, and adjust the "optimal distance" between the scanner and the scanned object according to the distance indicator bar.



– IR exposure is set to default automatic exposure (for large objects, human bodies or faces, automatic mode can be used). If the exposure effect is not satisfactory in automatic mode, you can set it to manual

exposure and adjust the picture to achieve normal exposure to obtain better results.

2. Click "Start" to start real-time scanning. During the scanning process, while maintaining a relatively fixed distance, carefully and slowly move the scanner towards the unscanned part of the object, and continue to ensure normal exposure and "optimal distance".
3. Turn on the "Quality Map" mode, continue scanning according to the interface prompts until all point clouds of the scanned object turn green, indicating that all model data has been acquired and the model quality is high.



Click "Complete" to end the scan and enter the processing stage.

Processing Page



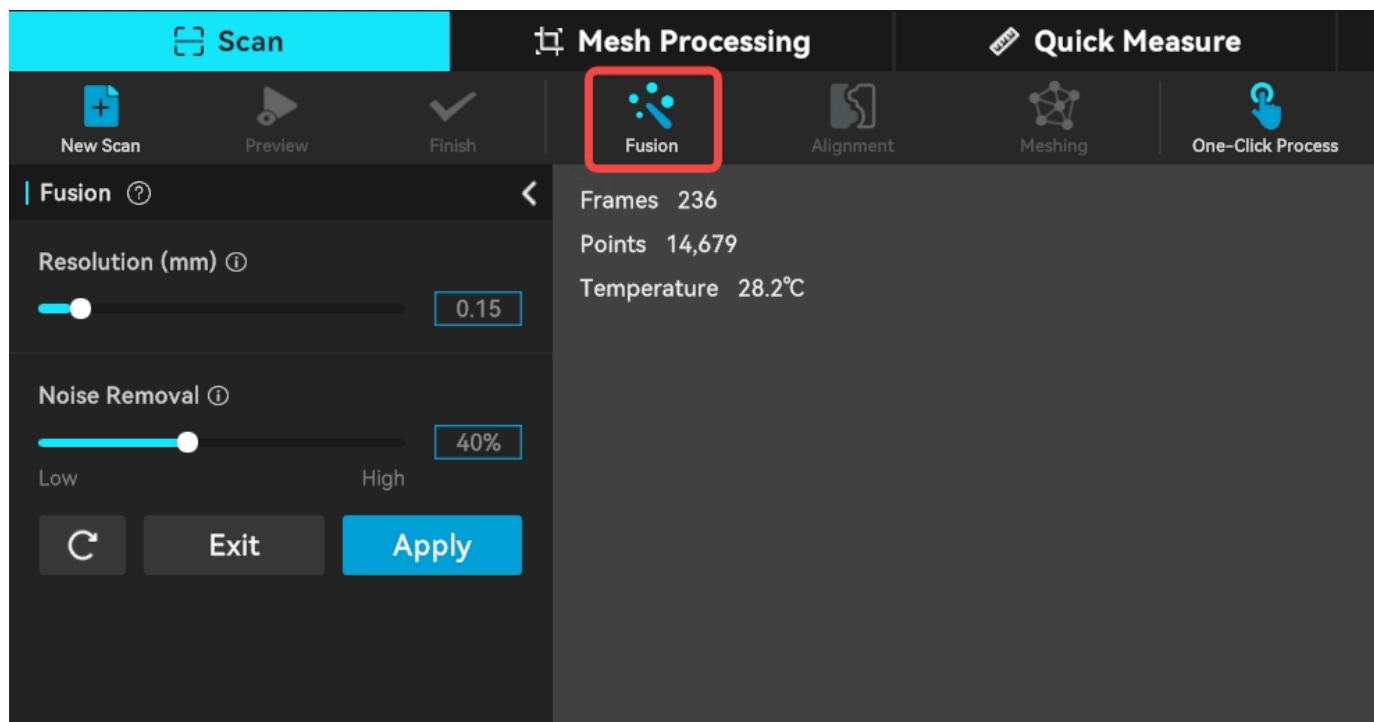
After scanning is completed, it supports one-click process or manual processing. The sequence of manual processing is fusion - meshing - color mapping.

Before fusion, you can edit the point cloud (delete redundant point clouds). After fusion, you can choose to continue editing the point cloud or use it for alignment (only available when there are two or more fused point clouds in the project panel).

Point Cloud Fusion

Convert point cloud data into a mesh model. There are two ways to achieve this function. Automatic mode ("one-click process" under "Scan") will use default parameters for fusion; Manual mode allows manual setting of resolution, noise removal, and whether to remove markers.

The smaller the resolution, the better the details, but it requires more memory and processing time.

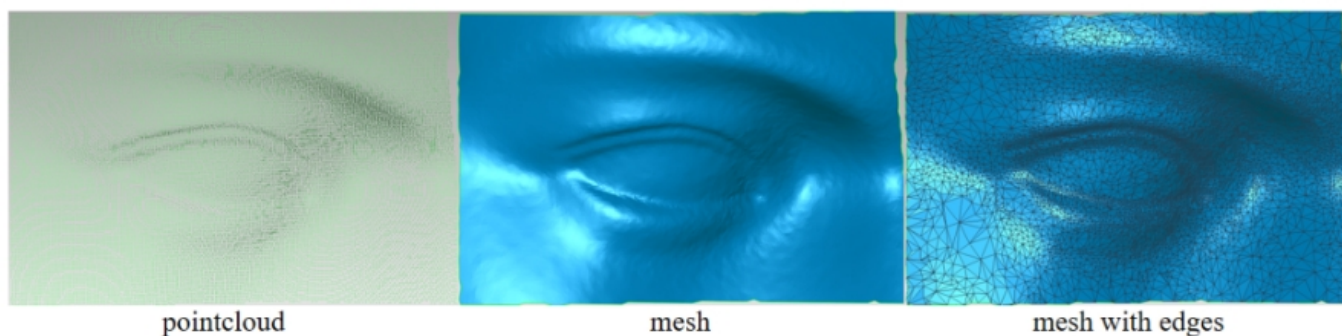


Noise Removal: The lower the noise reduction level, the more data can be retained.

Remove Markers: This function is only displayed when are set in markers tracking mode. It is used to remove markers and fill the corresponding holes. Removing markers will optimize the mesh for the identified marker positions.

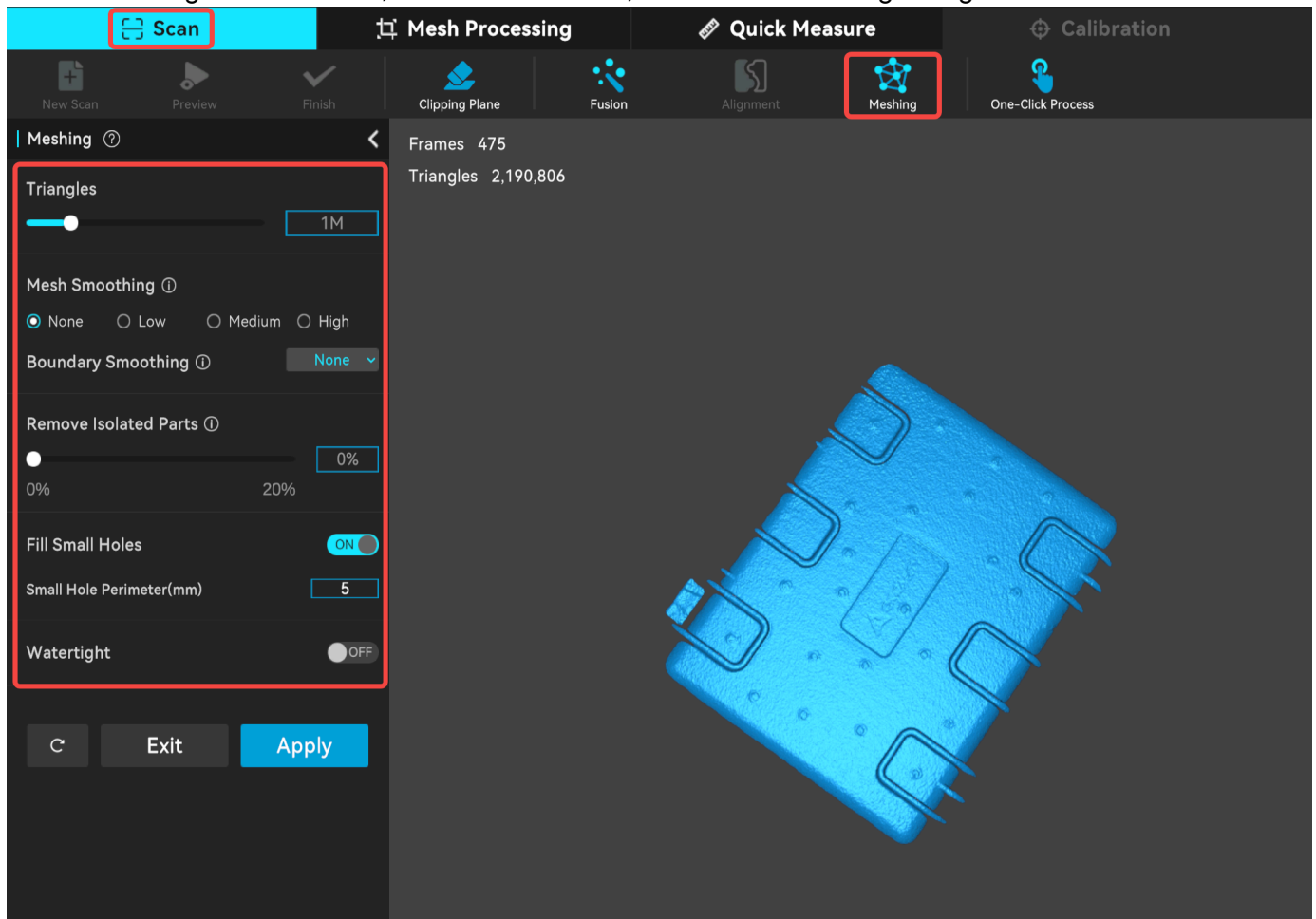
Meshing

The meshing operation converts pointcloud data into a triangular mesh model.



During the meshing process, you can manually configure the number of the triangles, adjust the mesh smoothness, set the percentage for removing isolated parts, and choose whether to remove spikes, fill holes, and make the model watertight.

The more triangles in the mesh, the finer the model, but it also takes longer to generate the mesh.



- ▶ **Triangles:** The higher Triangles is set, the higher the degree of detail restoration of the model.
- ▶ **Mesh Smoothing:** The higher the smoothness level, the smoother the mesh.
- ▶ **Boundary Smoothing:** Smooths the model and the boundaries of holes. The higher the smoothness level, the smoother the mesh.
- ▶ **Remove Isolated Parts:** Eliminates isolated components in the model. The larger the value, the more faces will be removed.
- ▶ **Fill Small Holes:** The hole-filling mode will automatically fill holes in the model.
- ▶ **Watertight:** The closing mode will automatically generate the missing surfaces of the model and create a geometrically closed 3D model. The non-closing mode will not automatically generate missing surfaces.

Point Cloud/Mesh Editing

The main window is mainly used to display point clouds or models. When the cursor enters the 3D model preview area, you can operate point clouds/mesh models, as follows

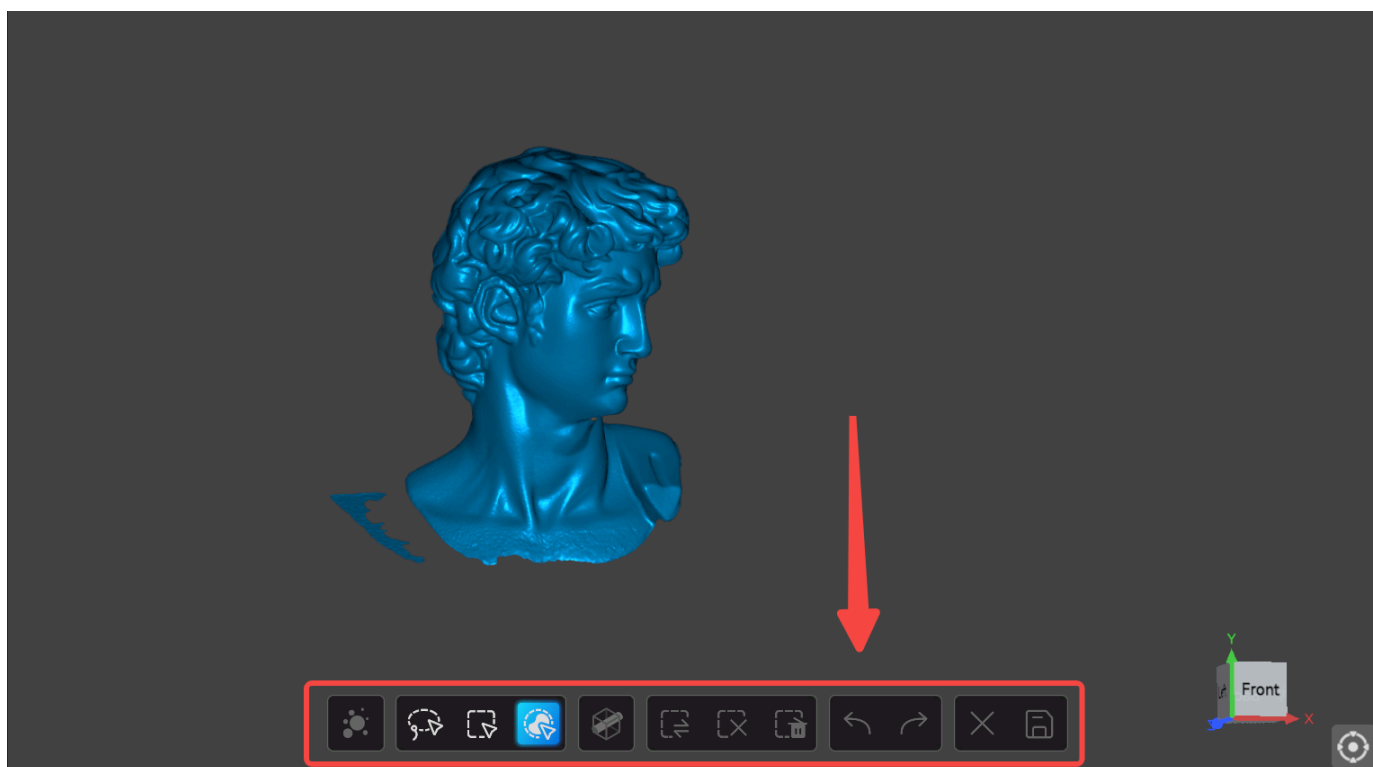
Button	Operation
Right mouse button	Click to select point cloud

Button	Operation
Left mouse button	Hold and drag to pan the point cloud
Middle button pressed	Hold and drag to rotate the point cloud
Middle mouse button scroll	Scroll to zoom in/out the point cloud

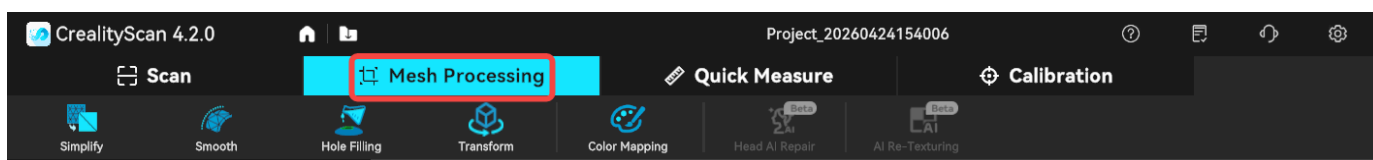
After the scan is completed, the, original point cloud and fused point cloud can be edited and processed. The editing toolbar is located on the right side of the main area.

Icon	Function Description
	Penetrate Selection: Supports selecting the mesh behind the selected area. For the raw data, selection always has a penetrate selection effect by default, so this option cannot be turned off.
	Lasso Select: Selects an area by dragging the mouse.
	Rectangle Select: Selects an area in a rectangular shape.
	Connected Component: Select all connected points in the clicked area.
	Invert Selection: Automatically selects the unselected area (opposite of the current selection).
	Unselect: Cancels the current selection.
	Delete Selection: Removes the selected part.
	Undo: Reverts the last deletion operation.
	Redo: Restores the last undone deletion operation.
	Cancel Edit: Exits the editing mode without saving changes.

Icon	Function Description
	Saved Edit: Saves the editing changes and exits the editing mode.



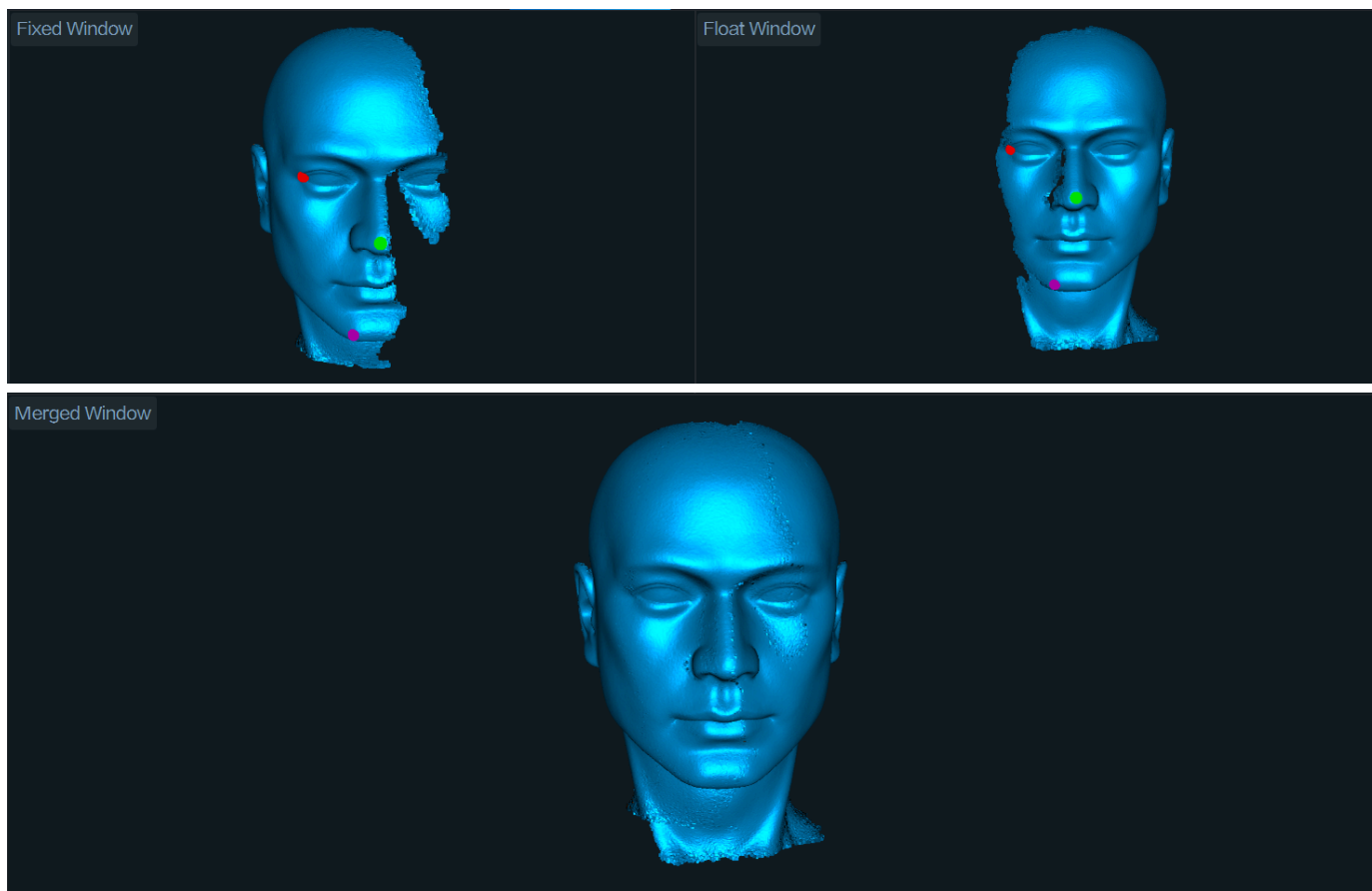
Similarly, the meshed model can also be edited and processed (click the "Edit" button in the upper toolbar to perform "Mesh Simplification", "Mesh Smoothing", or "Hole Filling" operations.)



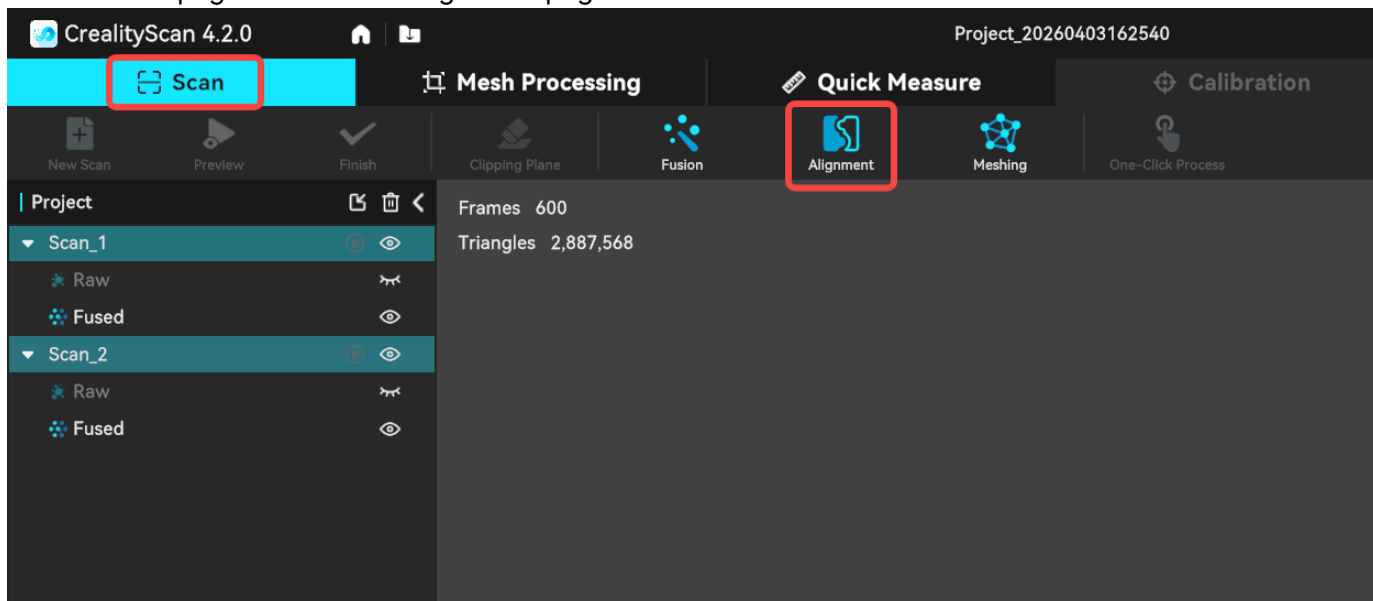
Multi-project Alignment

Each scan can capture only a portion of the object, and then you can use the Multi-Project alignment feature to combine them into a complete model. This is extremely helpful for scanning larger objects.

After completing the first scan and performing pointcloud fusion, you can start a new scan by selecting "New Project." On the page for Multi-Project alignment, you can automatically or manually align the fused point clouds.



When there are at least two projects in the multi-project panel, click "Alignment" in the toolbar at the top of the "Scan" page to enter the Alignment page.

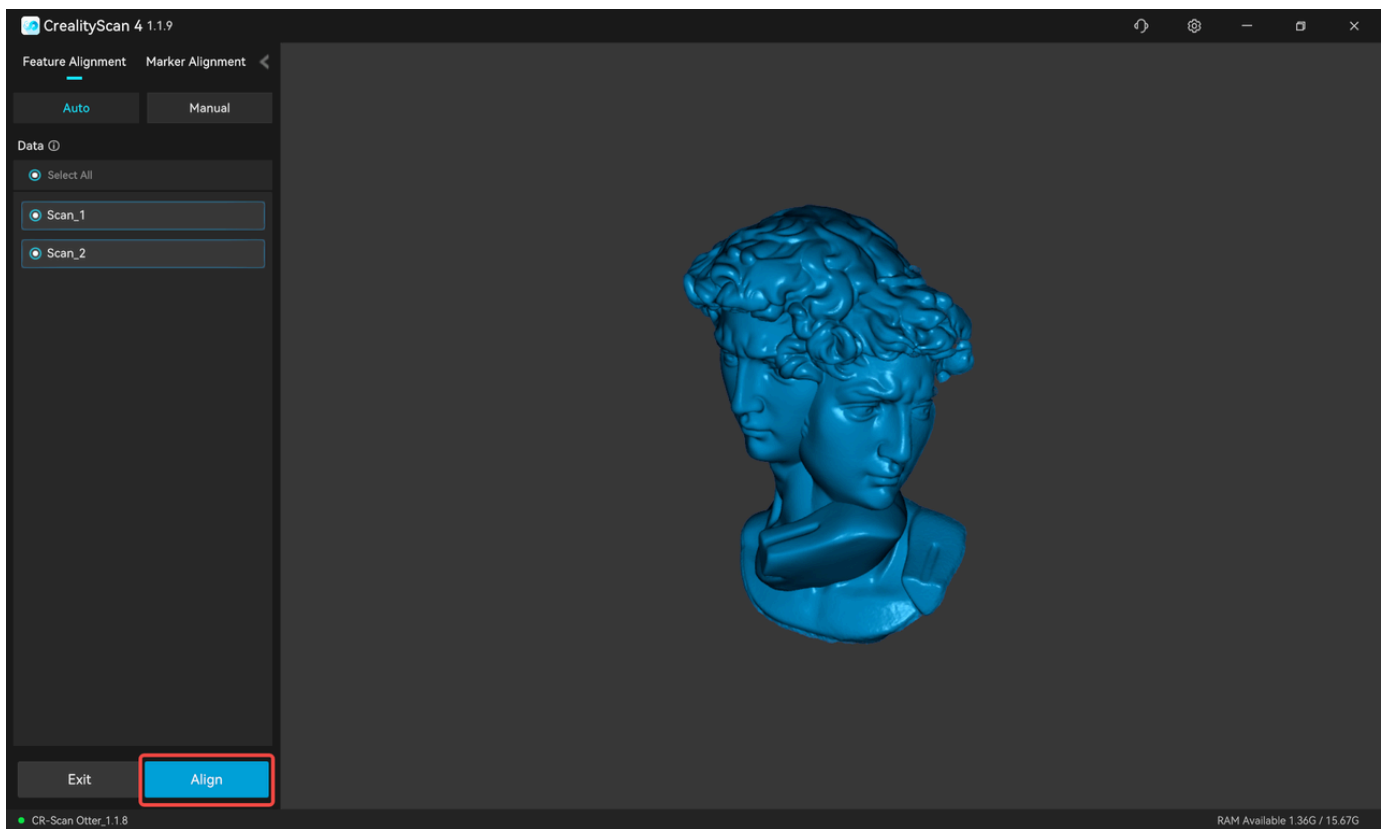


There are two types of align methods: "feature alignment" and "marker alignment", and users can switch the alignment method in the upper part of the left sidebar .

1. Feature alignment

Feature alignment provides both automatic and manual modes.

① Automatic alignment requires a sufficiently large common feature area between models.

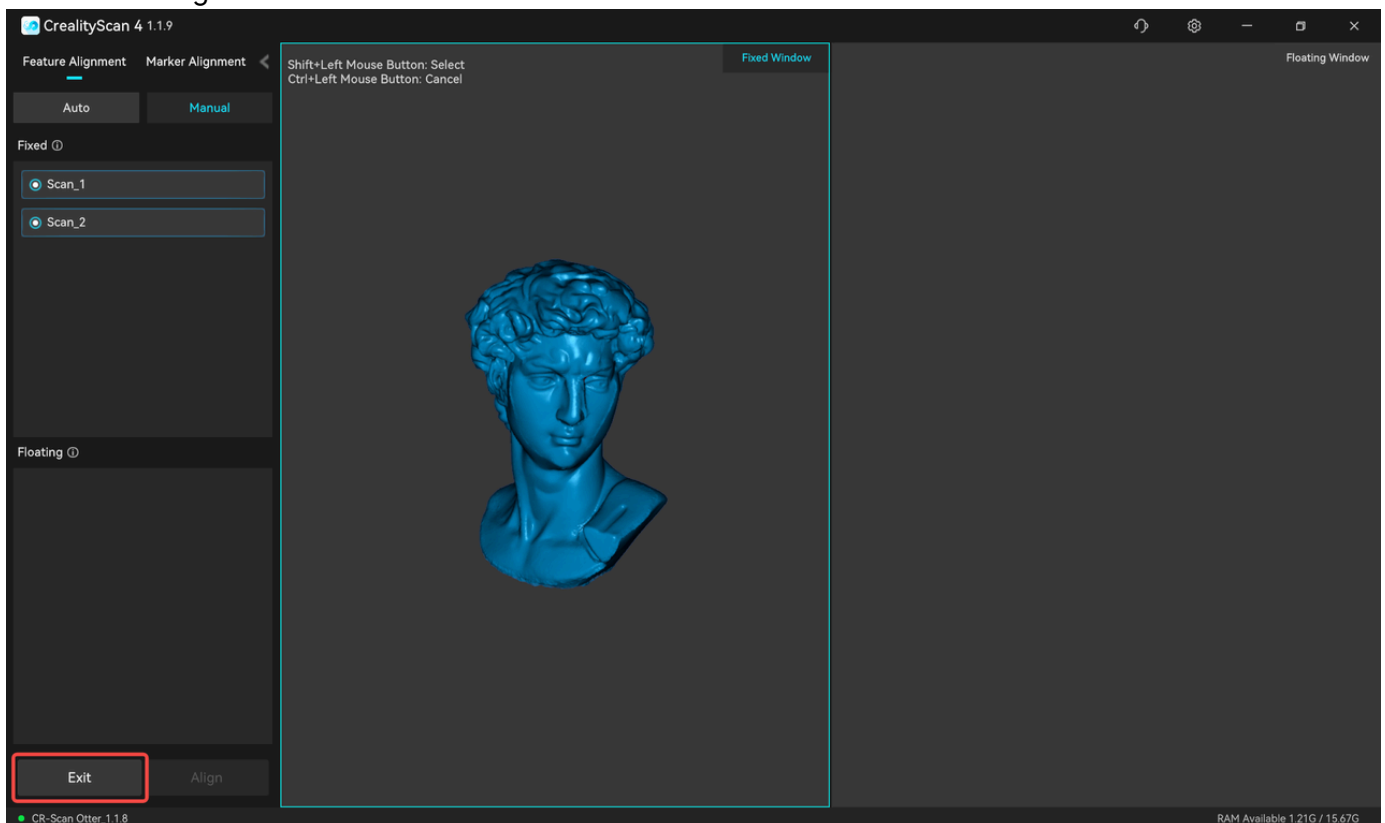


It is generally more recommended to use Manual alignment.

On the left is the model list. In automatic Alignment mode, select two sub-projects to be spliced in the model list, then click "preview" to start alignment. After Alignment is completed, click "apply" to complete the alignment.

② Manual alignment

Manual alignment requires at least three pairs of corresponding points on the models of the fixed window and the floating window.



In the main area, the left side is a fixed window, and the right side is a floating window. In the manual Alignment mode, select a sub-project from each of the two lists on the left and display them in the corresponding fixed window and floating window, then select at least three pairs of matching points on the models in the fixed window and floating window. Finally, click "preview".

After confirming the Alignment result, click "apply" to save the change. User can continue to align one by one, or click "exit" to leave the Alignment page.

2. Marker Alignment

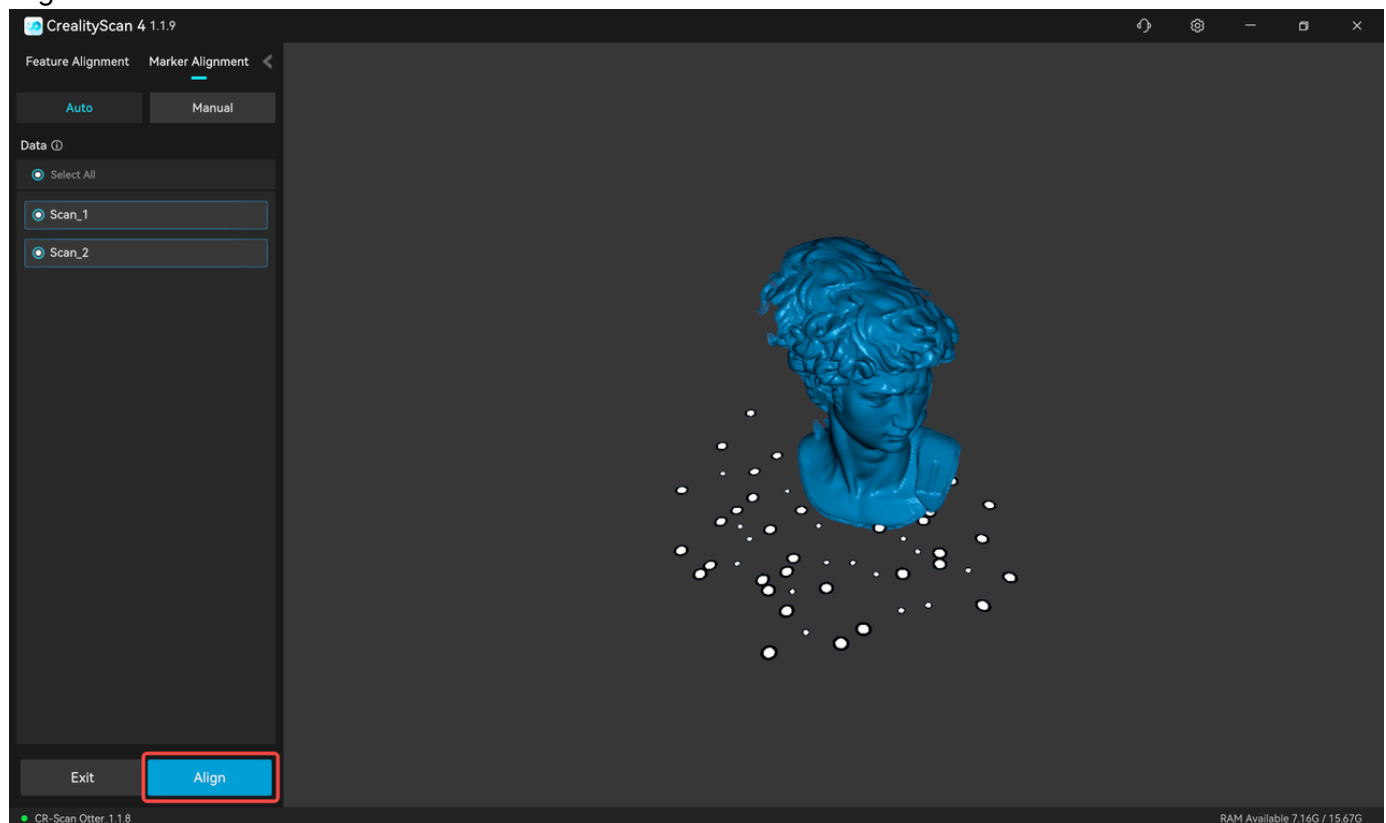
If you use "Marker" scanning mode, "Marker Alignment" is also available.

Marker alignment provides Auto and Manual modes.

- ① Auto alignment requires enough overlapping marker areas between models.
- ② Manual alignment requires at least three corresponding markers on each model.

Auto alignment

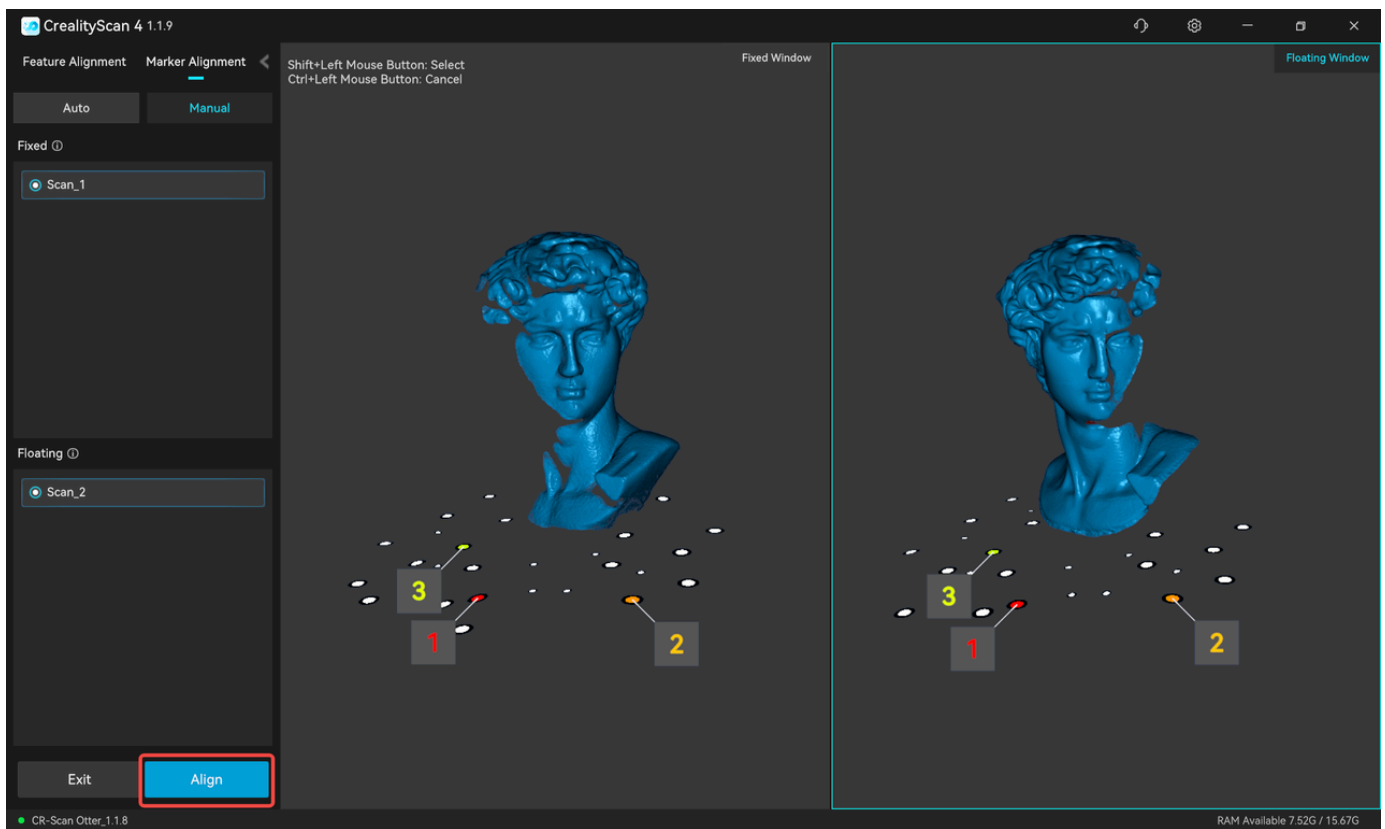
On the left is the model list. In auto alignment mode, select the two target subprojects in the model list, and then click "preview" to start aligning. After the alignment is completed, click "apply" to complete the alignment.



Manual alignment

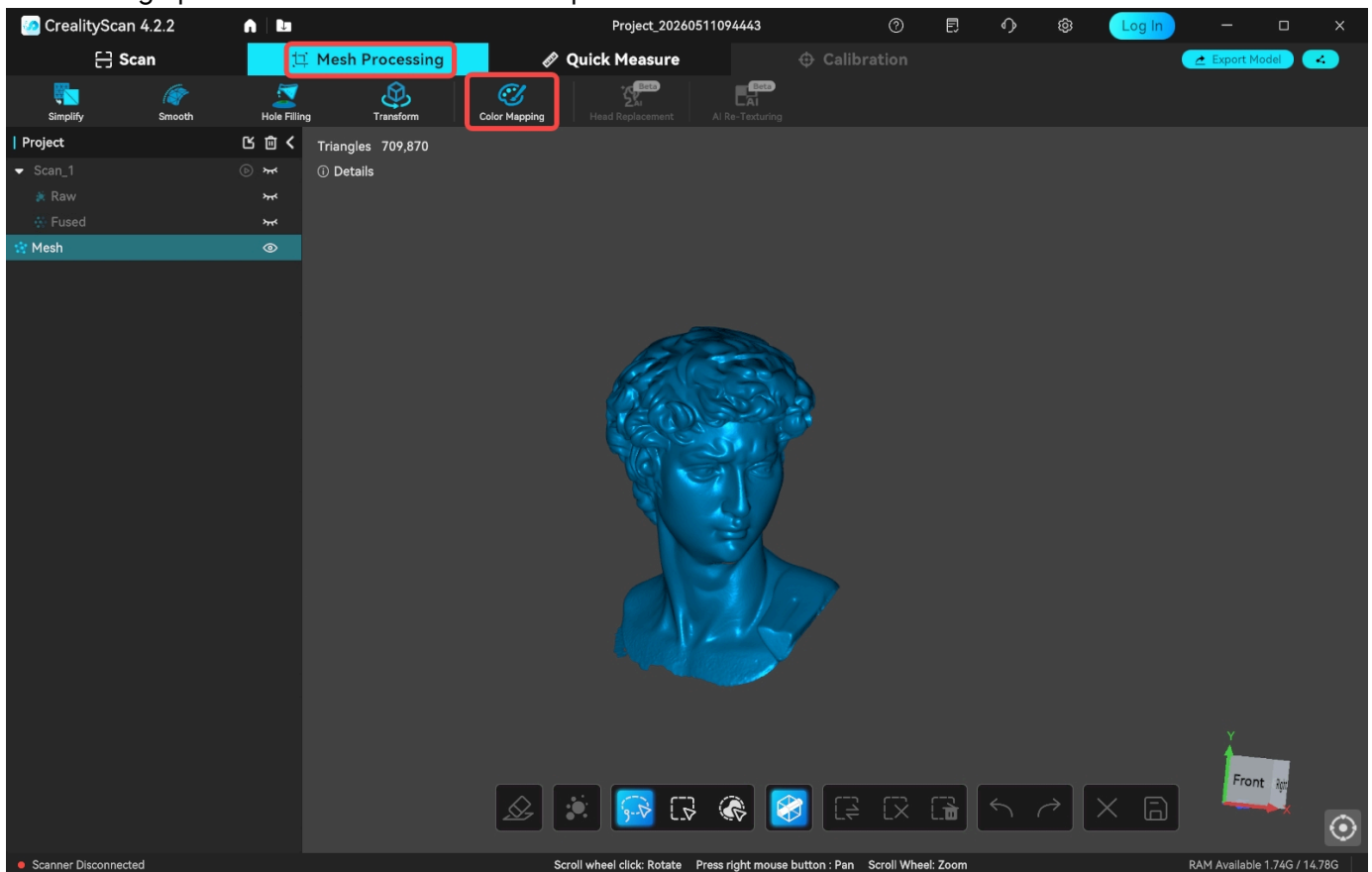
In the main area, the fixed window is on the left and the floating window is on the right. In manual alignment mode, select a subproject in the two lists on the left and display it in the corresponding fixed window and floating window, and then select at least three pairs of markers on the model in the fixed window and floating window. Finally click "preview".

After confirming the alignment effect in the window, click "apply" to complete the alignment.

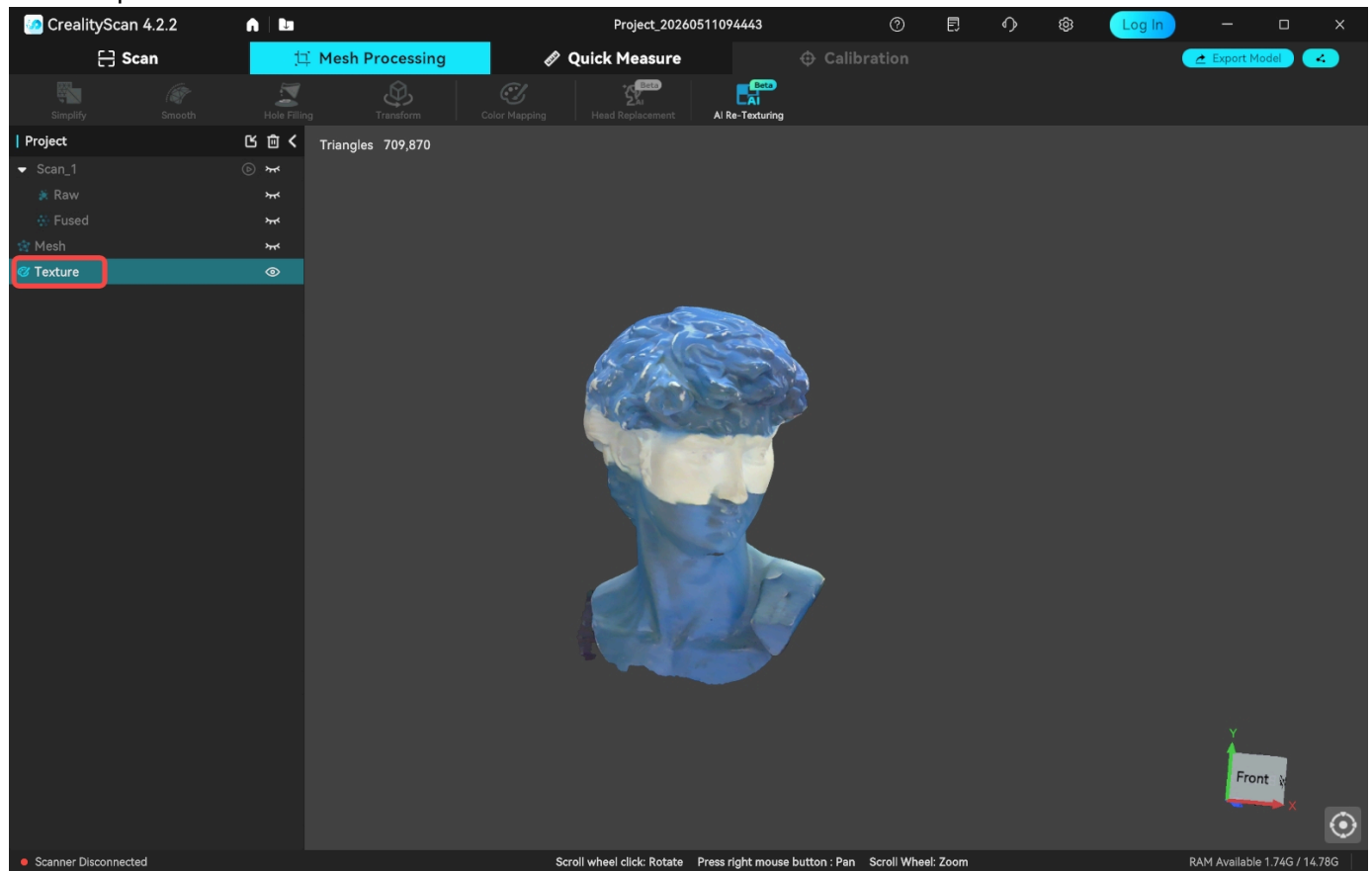


Color Mapping

Match image pixels on the mesh model to improve texture and color resolution.



The completed effect :



AI Re-Texturing

Supports capturing high-definition images using a mobile phone or DSLR camera, and then generating a more detailed textured model after import. For a detailed guide, please refer to [AI Re-Texturing](#).

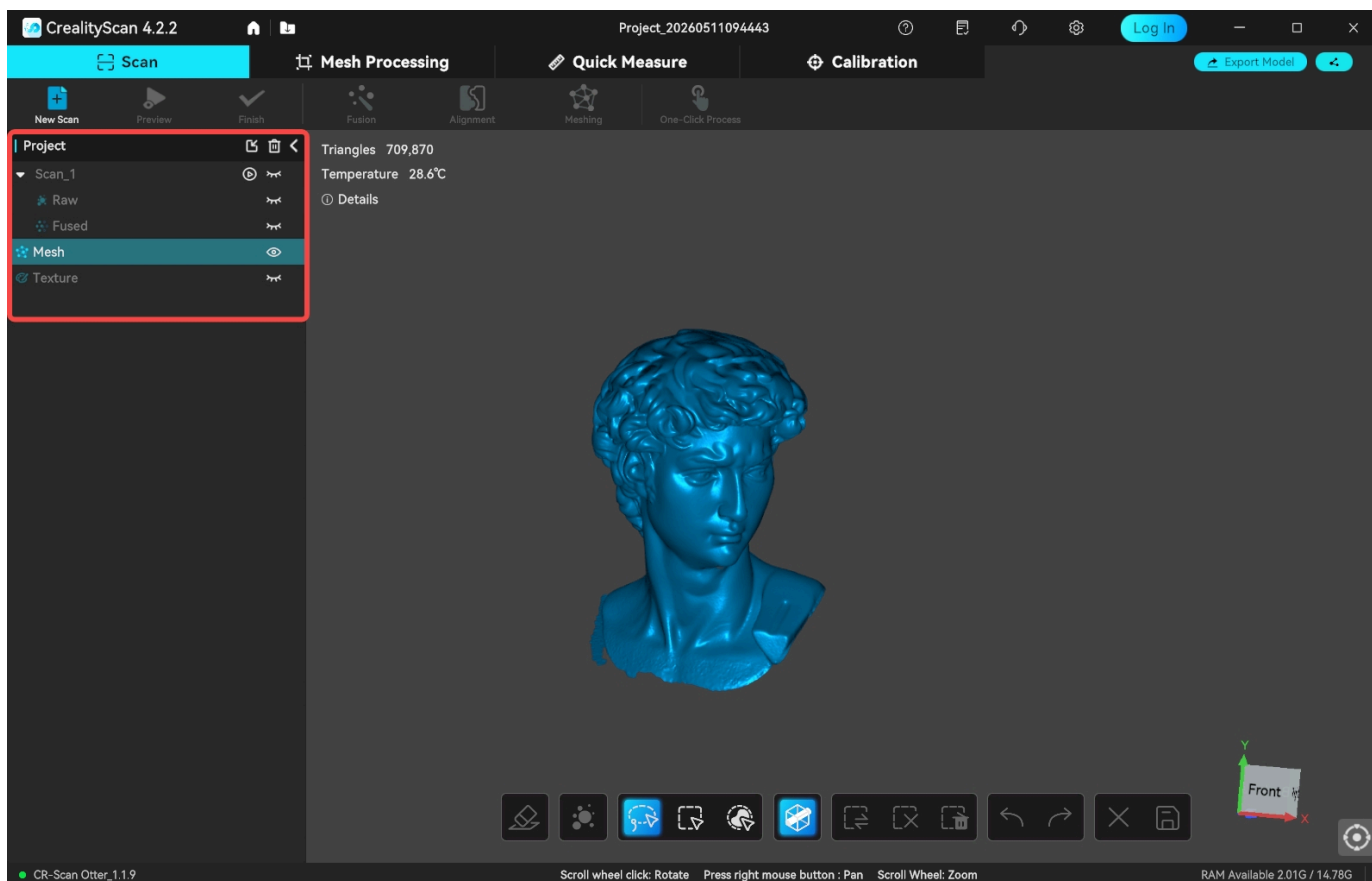
Gaussian Rendering

Gaussian Splatting is a 3D rendering technology that enhances the realism and visual quality of scanned models. Based on this technology, CrealityScan offers two features: Native Scan Gaussian Reconstruction and Re-Texturing Gaussian Generation, to produce Gaussian Splatting effects.

AI Body Completion

No need to scan your whole body. Just upload one clear front-facing photo, and AI will intelligently generate a complete body model and seamlessly merge it with scanned high-precision body data to create a full 3D digital avatar in one click.

The multi-project panel



1. Data states display

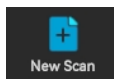
The multi-project panel on the left shows the currently open projects/scans. Each sub-project unlocks up to four data states with the processing stage, namely **raw**, **fused**, **mesh**, and **texture**. Users can right-click to delete the corresponding data or rename the project.

2. Continue scanning



Click to continue scanning.

3. Create New Scan/Import historical projects

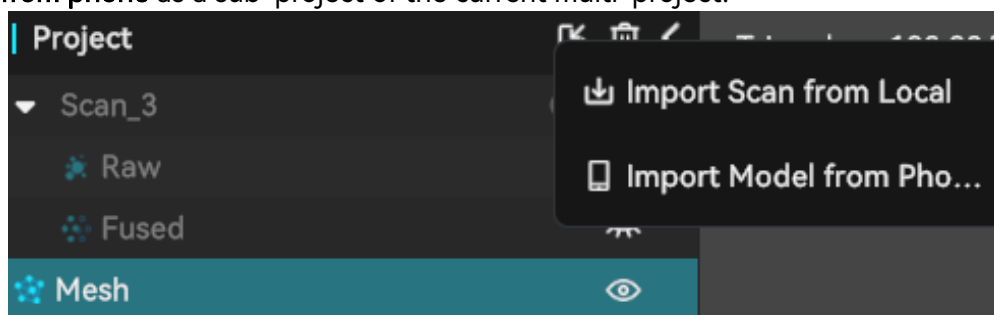


Click "New Scan" to create a new scan as a subproject of the current multi-project.

Click the first icon to the right of "Project". This feature supports:

Import Scan from Local as sub-projects of the current multi-project.

Import Model from phone as a sub-project of the current multi-project.



In the upper function bar, manual processing is recommended. The processing order is fusion – meshing – color mapping . "Point cloud fusion" is under "Scan", and "meshing" and "Color mapping" are under "Process".

Export/Share 3D model



Click the upper right corner icon of the software to **Export completed 3D models**. Fusion supports exporting in **.asc and .ply** formats. Mesh supports exporting in **.stl, .ply, and .obj** formats. Texture supports exporting in **.ply and .obj** formats.

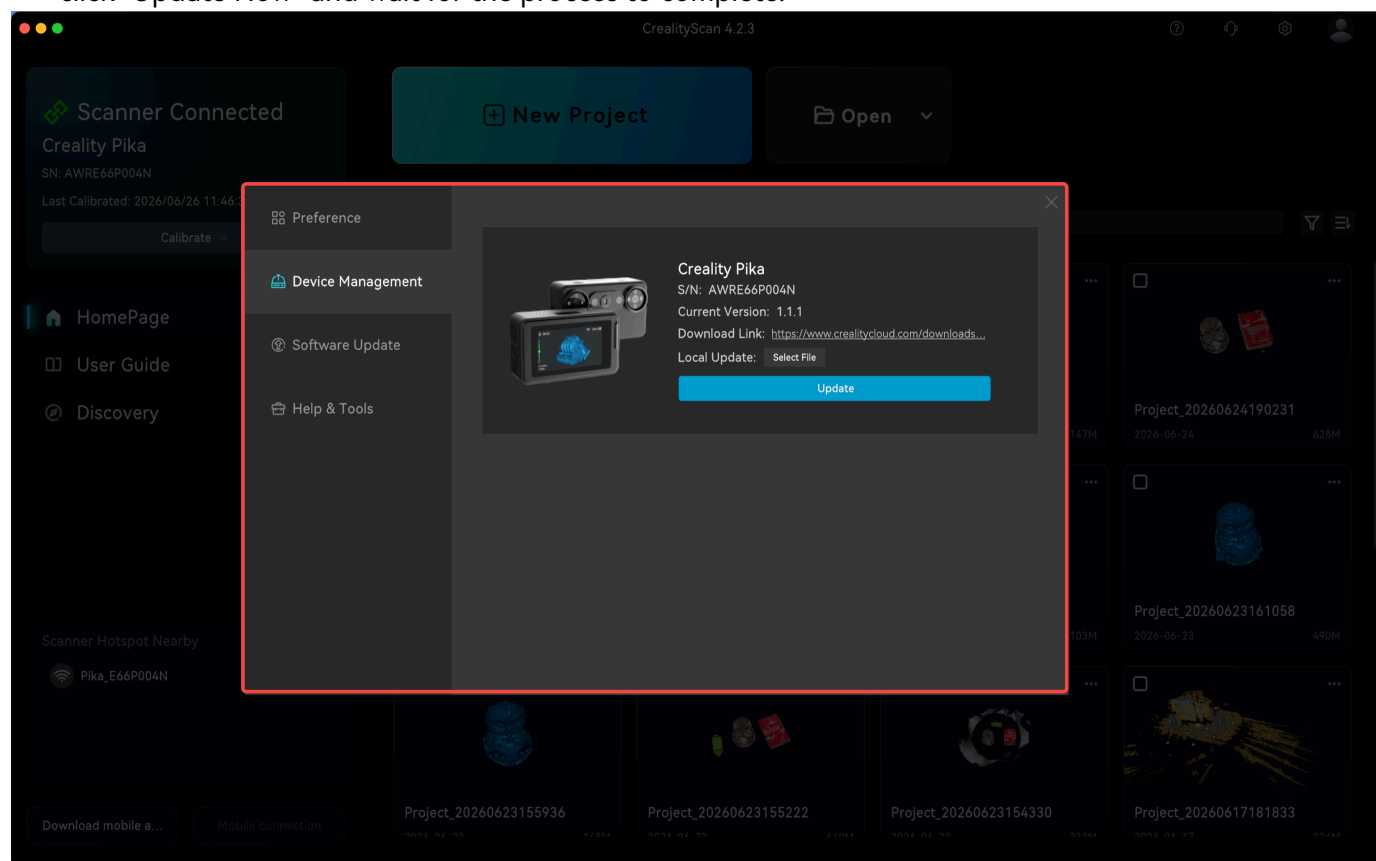


Click the upper right corner icon of the software to share the processed 3D model to the third party software, like Creality slicing software for direct printing.

Firmware Upgrade

Firmware upgrade is divided into automatic detection and upgrade and manual firmware upgrade.

- ▶ After the scanner is successfully connected, if the software has an internet connection, it will automatically detect whether the scanner's firmware version needs an update. You can proceed with the update after confirming it manually.
- ▶ To update the firmware manually, navigate to **Settings > Firmware Update** in the top right sidebar. Download the corresponding latest firmware from the provided download link, click "Select File", then click "Update Now" and wait for the process to complete.



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